JVM Architecture and Runtime Model

How your Java code comes to life — from source file, through bytecode, to a running program managed by the Java Virtual Machine.

TEKsystems Global Services

WORKFORCE DEVELOPMENT

WHY UNDERSTANDING THE JVM MATTERS

The JVM is the foundation of every Java application — understanding it builds applications that are reliable, efficient, and production-ready.

Better Performance

Tune memory, GC and threads.

Faster Troubleshooting

Diagnose leaks, crashes, issues quickly.

Reliability at Scale

Handle enterprise workloads confidently.

Efficient Resources

Optimize CPU, memory and I/O.

Stronger Foundation

Write better code, grow expertise.

Java Source

You write .java code.

Compiler

javac checks & compiles it.

Bytecode

Platform-independent .class.

JVM

Loads, verifies, executes.

OS

Program runs, output appears.

- Real-world impact: handle millions of requests, avoid outages and downtime, deliver a consistent user experience, and scale with confidence.

KEY TAKEAWAY The JVM is the bridge between your code and the operating system — master it to build software that performs and scales.

MODULE 1 LEARNING OBJECTIVES

By the end of this module, you will understand the core components, memory model, and execution flow of the JVM.

- Explain the role of the JVM in the Java platform and its importance in application execution.
- Identify and describe the major components of the JVM and how they work together.
- Understand the compilation process and the structure of Java bytecode.
- Describe the class loading mechanism and the lifecycle of a class in the JVM.
- Explain the JVM memory model, including Heap, Stack, Metaspace and Method Area.
- Understand garbage collection concepts and object lifecycle management.
- Analyze JVM performance factors and common memory-related issues.
- Use JVM tools and monitoring concepts to diagnose and troubleshoot runtime problems.

KEY TAKEAWAY This module builds the foundation every later module — memory, GC, performance — depends on.

ENTERPRISE SCENARIO: DIAGNOSING A PRODUCTION MEMORY ISSUE

A customer management application is experiencing high memory usage and frequent OutOfMemoryError in production, impacting users.

SYMPTOMS OBSERVED

Heap Usage 95%+ Frequent Full GC taking several seconds.

Slow Threads OutOfMemoryError appears in application logs.

Timeouts Service interruptions; users see errors and slow responses.

Business impact: Lost transactions, unhappy customers, SLA breaches.

KEY TAKEAWAY Learning outcome: understand how JVM knowledge helps identify the root cause and resolve memory issues in production.

HOW JVM KNOWLEDGE HELPS

Turning a production outage into a diagnosed, fixed, and prevented issue.

- Understand the JVM memory model: Heap, Stack, Metaspace, etc.
- Identify the memory leak using heap dumps and memory analysis tools.
- Analyze object allocation and garbage collection behavior.
- Tune JVM parameters for optimal memory usage.
- Implement code fixes and prevent future memory issues.
- Restore application stability and improve performance.

Business Impact: faster issue resolution, improved application stability, better user experience, and reduced operational costs.

KEY TAKEAWAY Deep understanding of JVM architecture, memory management and runtime behavior is essential to resolve real-world production issues.

WHAT IS JAVA?

A high-level, class-based, object-oriented language designed by Sun Microsystems (now Oracle), released in 1995.

Simple

- Easy to learn, clean syntax, reduces complexity.

Object-Oriented

- Encapsulation, Inheritance, Polymorphism, Abstraction.

Platform Independent

- Write Once, Run Anywhere (WORA) via the JVM.

Secure

- Built-in security, automatic memory management, no pointer arithmetic.

Robust

- Strong memory management, exception handling, multi-threading.

High Performance

- JIT compilation and efficient runtime optimizations.

KEY TAKEAWAY These six characteristics are why Java has remained a dominant enterprise language for three decades.

HOW JAVA WORKS

The same journey from source code to execution repeats throughout this module.

WRITE ONCE, RUN ANYWHERE (WORA)

- Windows
- Linux
- macOS
- Unix
- Others

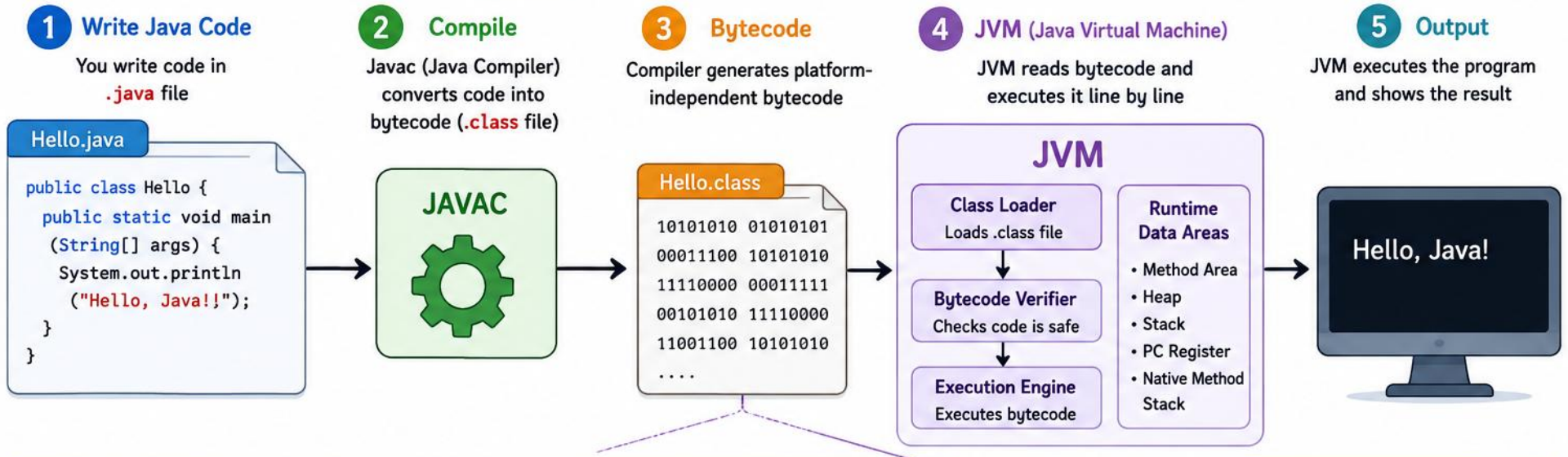
WHERE IS JAVA USED?

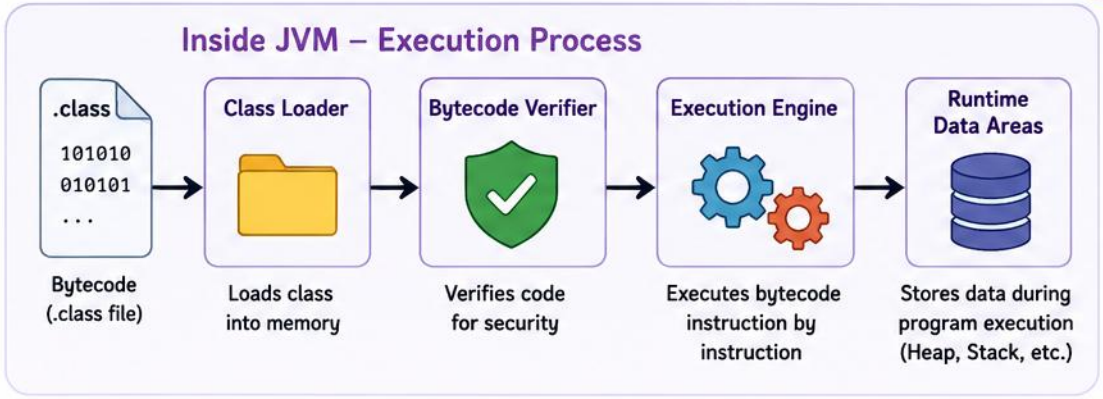
- Enterprise & Cloud
- Mobile (Android)
- Big Data
- Games & IoT

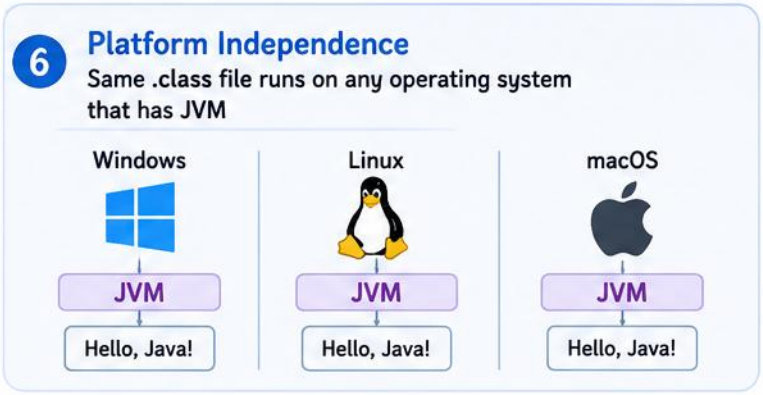
WORA & FULL PIPELINE Write Once → Compile Once → Run Anywhere (WORA). End-to-end: Class Loader → Runtime Data Areas → Execution Engine (interpreter/JIT, JNI for native calls) → Result → Terminate — the 8-step flow behind every run above.

KEY TAKEAWAY Bytecode is the platform-independent middle step that makes every one of these use cases possible.

How Java Works







Summary Write Code (.java) → Compile (javac) → Bytecode (.class) → Run on JVM → Output

Java source code is compiled into bytecode, and JVM executes it.

JAVA PLATFORM OVERVIEW

A rich, secure, and reliable environment for building, deploying, and running enterprise applications.

Platform Independent
Write once, run anywhere.

Secure
Built-in security & best practices.

Robust
Memory mgmt, exceptions, typing.

High Performance
JIT compiler, optimizations.

Large Ecosystem
Rich libraries, frameworks, tools.

JAVA SE — WHAT IT'S MADE OF

Dev Tools & APIs

- javac, javadoc, JConsole, JMC, and more.

Core Libraries (API)

- Collections, IO/NIO, Concurrency, JDBC, Networking.

Java Virtual Machine

- Class Loader, Runtime Areas, Execution Engine, GC, Security Mgr.

Operating System

- Windows, Linux, macOS, Unix, and others.

KEY TAKEAWAY The JVM is only one layer of the Java Platform — tools, libraries, and the OS all sit around it.

JAVA PLATFORM EDITIONS

Three editions of the platform, sized for three very different classes of application.

Java SE (Standard Edition)

- Foundation for desktop apps, small servers, core enterprise services.
- Core Java APIs, development tools.
- Desktop and server applications.

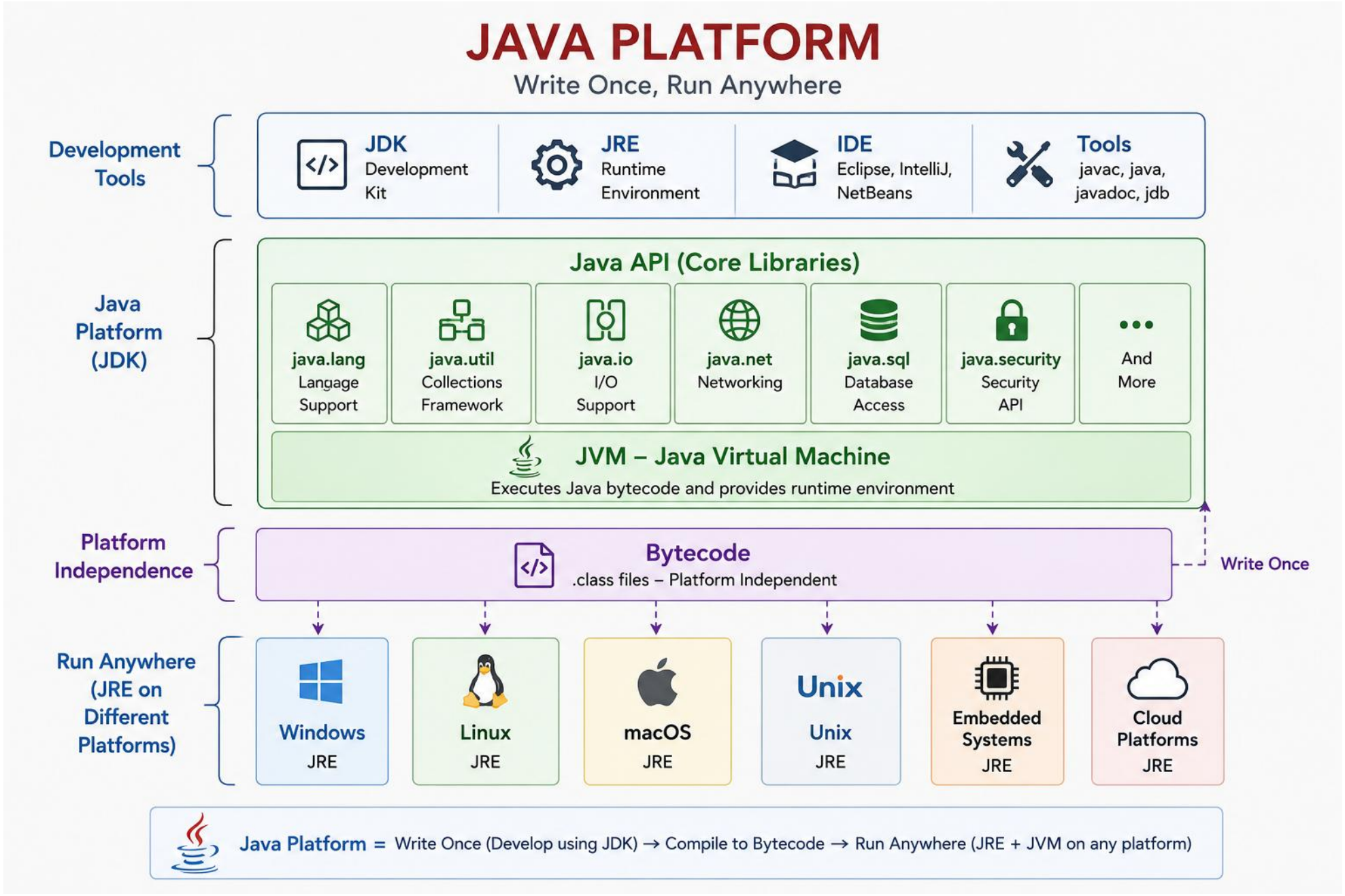
Java EE (Enterprise Edition)

- Built for enterprise apps with distributed-system APIs.
- Web, EJB, JMS, JPA, Security.
- Scalability & high availability — now evolved as Jakarta EE.

Java ME (Micro Edition)

- For resource-constrained and embedded devices.
- Compact APIs optimized for small devices.
- IoT and embedded use cases.

KEY TAKEAWAY This bootcamp focuses on Java SE — the foundation every other edition builds on.



JDK VS JRE VS JVM

Three essential components of the Java Platform, each with a specific purpose.

JDK

- Complete dev environment to build, test, package apps.
- Includes JRE + Dev Tools (javac, java, javadoc, jar, jdb, jlink).
- Used by: Developers.

JRE

- Libraries, JVM, and components required to run apps.
- Includes JVM + Core Libraries + supporting files.
- Used by: End users, to run Java applications.

JVM

- The heart of Java — executes bytecode, provides platform independence.
- Loads, verifies, executes bytecode; manages memory.
- Used by: JRE and Java applications.

IN PRACTICE javac Hello.java needs the JDK. The resulting Hello.class then runs on any machine with just a JRE — the JVM inside it loads, verifies, and executes the bytecode to print "Hello Java!".

KEY TAKEAWAY JDK is a superset containing JRE + dev tools; JRE provides everything to run an app; JVM is the engine that executes bytecode.

JDK vs JRE vs JVM

Understanding the Relationship and Key Differences

JDK
Java Development Kit

JDK is a superset of JRE.
It provides all the tools and libraries required to develop, compile, debug, and run Java applications.

JDK = JRE + Development Tools

JRE (Runtime Environment)

- JVM (Java Virtual Machine)
- Java Class Libraries

Development Tools

- javac (Compiler)
- jdb (Debugger)
- jar (Archiver)
- javadoc (Documentation)
- keytool (Key Generator)
- and more

Used For: Development + Execution
(Build and run Java applications)

JRE
Java Runtime Environment

JRE provides the libraries, JVM, and other components to run Java applications.
It does not include development tools.

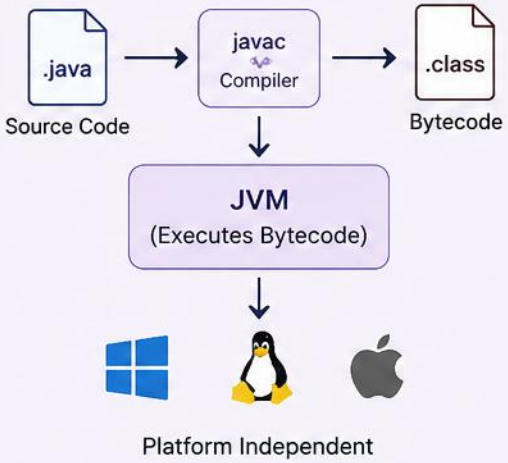
JRE

- JVM (Java Virtual Machine)
- Java Class Libraries & Other Libraries

Used For: Execution
(Run Java applications)

JVM
Java Virtual Machine

JVM is the core component that runs Java bytecode (.class files) and makes Java platform independent.



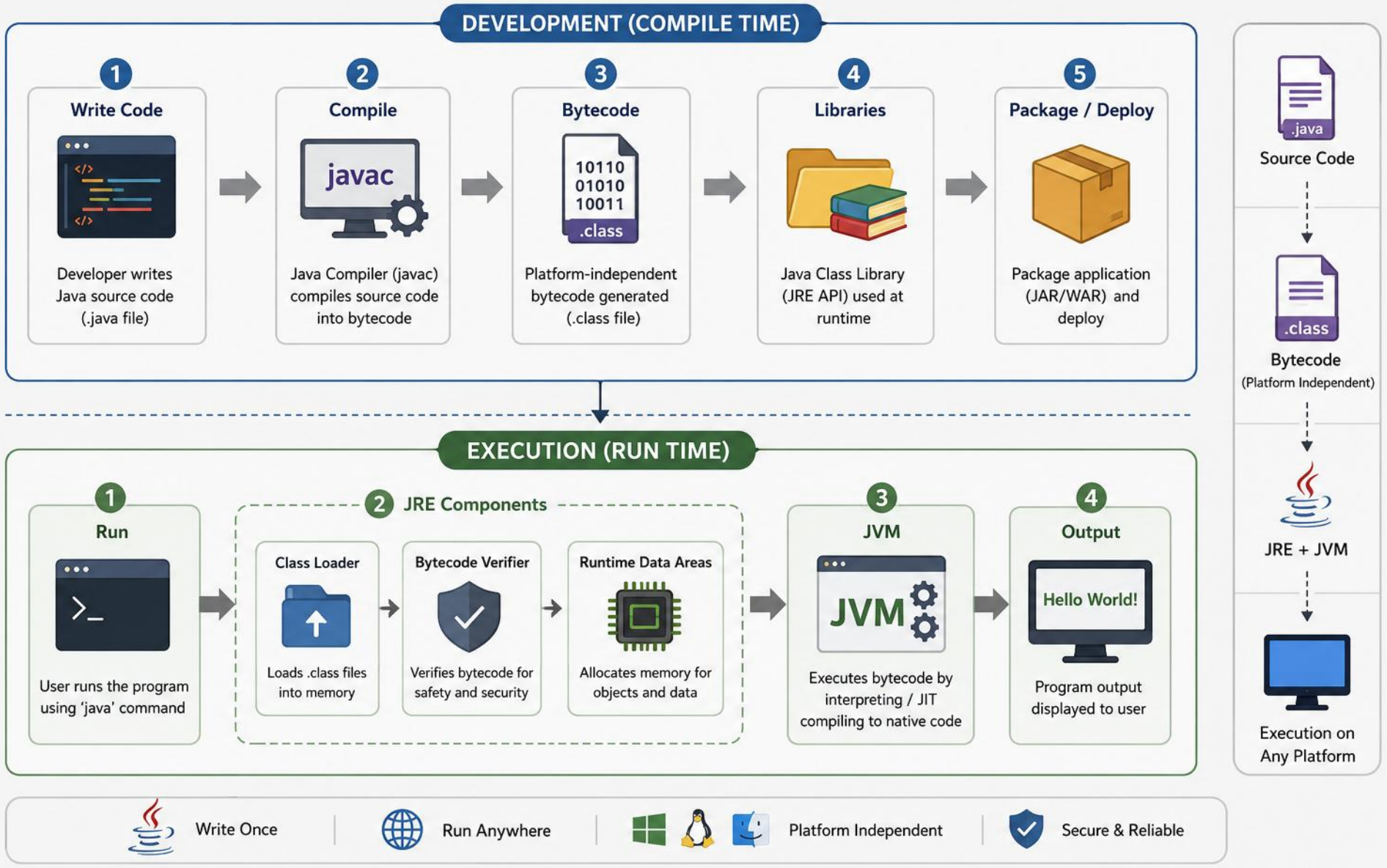
Used For: Execution Engine
(Runs bytecode on any platform)

Key Takeaway

- **JDK** = Everything needed to develop and run Java programs.
- **JRE** = Everything needed to run Java programs (no development tools).
- **JVM** = The engine that runs Java bytecode, part of JRE (and JDK).

JDK (Superset) ⊃ JRE (Subset) ⊃ JVM (Core Engine)

Java Development and Execution Workflow



TRY IT YOURSELF — EXERCISE 1: HELLO WORLD

Reinforces: the edit, compile, run workflow you just saw.

STEP	WHAT TO DO
Goal	Write, compile, and run a program that prints "Hello, JVM!"
File	Hello.java (in examples/module-01-exercises/)
Entry point	public static void main(String[] args) — the method the java launcher looks for
Compile	javac Hello.java produces Hello.class (bytecode)
Run	java Hello starts a JVM and executes that bytecode
Reference	exercises/exercise-01-hello-world.md

```
$ cd examples/module-01-exercises
$ javac Hello.java
$ java Hello  // prints: Hello, JVM!
$ javap -c Hello  // preview the bytecode — Exercise 8 covers this in depth
```

TRY IT NOW Complete Exercise 1 before continuing — see exercises/exercise-01-hello-world.md.

TRY IT YOURSELF — EXERCISE 2: PLATFORM INDEPENDENCE (WORA)

Reinforces: Write Once, Run Anywhere — the idea you just covered.

STEP	WHAT TO DO
Goal	Run an existing .class file and explain why no recompile is needed for another OS
File	Hello.class (already compiled in Exercise 1 — no source edit needed)
Command	java Hello — starts a JVM, loads Hello.class, runs main
Key idea	The same .class bytecode runs unchanged on any OS with a matching JVM
Deliverable	2–3 sentences: source vs. bytecode vs. JVM, saved as notes/wora-notes.md
Reference	exercises/exercise-02-wora.md

```
$ cd examples/module-01-exercises
$ java Hello
$ java Hello  // re-run with no javac — proves you're executing bytecode, not the .java file
$ java -version  // the same .class would run under this JVM on Windows, macOS, or Linux
```

TRY IT NOW Complete Exercise 2 before continuing — see exercises/exercise-02-wora.md.

JVM COMPONENTS OVERVIEW

The JVM is the runtime engine that executes Java bytecode and provides the runtime environment for Java applications.

KEY RESPONSIBILITIES

Execute Bytecode

Runs compiled Java bytecode.

Platform Independence

Write Once, Run Anywhere.

Manage Memory

Allocates & reclaims automatically.

Ensure Security

Secure execution environment.

Optimize Performance

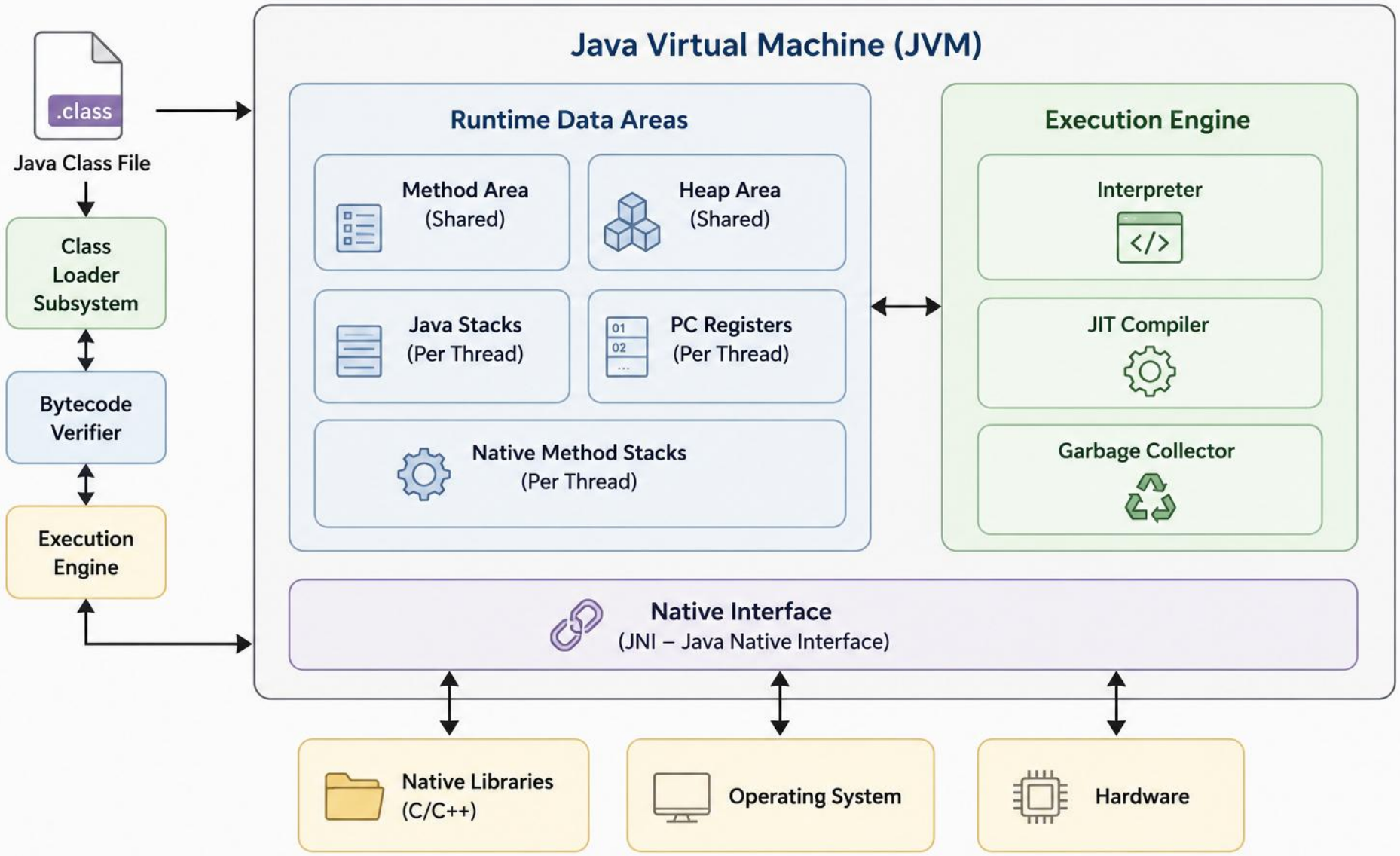
JIT compilation & tuning.

JVM ARCHITECTURE (HIGH LEVEL)

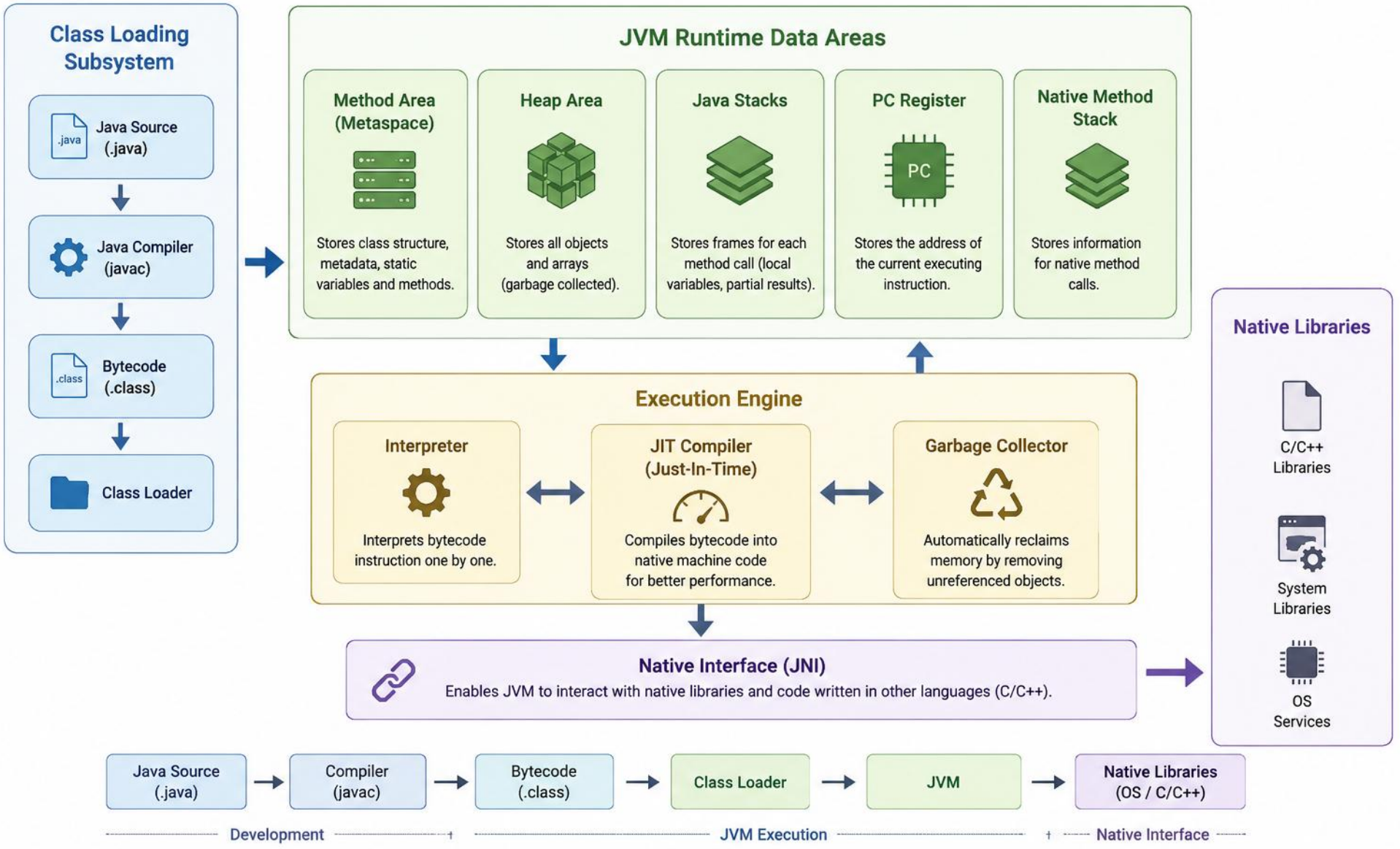
- Class Loader Subsystem: loads .class files into the JVM.
- Runtime Data Areas: Method Area, Heap, Java Stacks, PC Register, Native Method Stack.
- Execution Engine: interprets bytecode; JIT Compiler optimizes and executes code.
- Native Method Interface (JNI): lets Java code call native libraries and methods.
- Native Method Libraries: native libraries (C/C++, OS) used by the JVM and applications.
- Working Together: on every method call, these five components collaborate in sequence — Class Loader → Runtime Data Areas → Execution Engine → (JNI if native code is needed) → Output.

KEY TAKEAWAY JVM is the heart of the Java Platform — it provides platform independence, security, and performance through optimization.

JVM Components



Java Virtual Machine (JVM) Architecture



CLASS LOADER SUBSYSTEM

Responsible for loading .class files into the JVM, linking them, and preparing them for execution.

WHAT IT DOES

- Loads class files from various sources, links and verifies them, ensures they are safe and valid to execute, and prepares them for runtime usage.
- Class file sources: file system, JAR/WAR files, network locations, and databases.

CLASS LOADER ARCHITECTURE — DELEGATION MODEL

1 Bootstrap

- Loads core classes from rt.jar / Java 9+ modules.
- e.g. java.lang.*, java.util.*, java.io.*

2 Platform (Extension)

- Loads from jre/lib/ext or modules.
- e.g. javax.*, org.w3c.dom.*, org.xml.sax.*

3 Application (System)

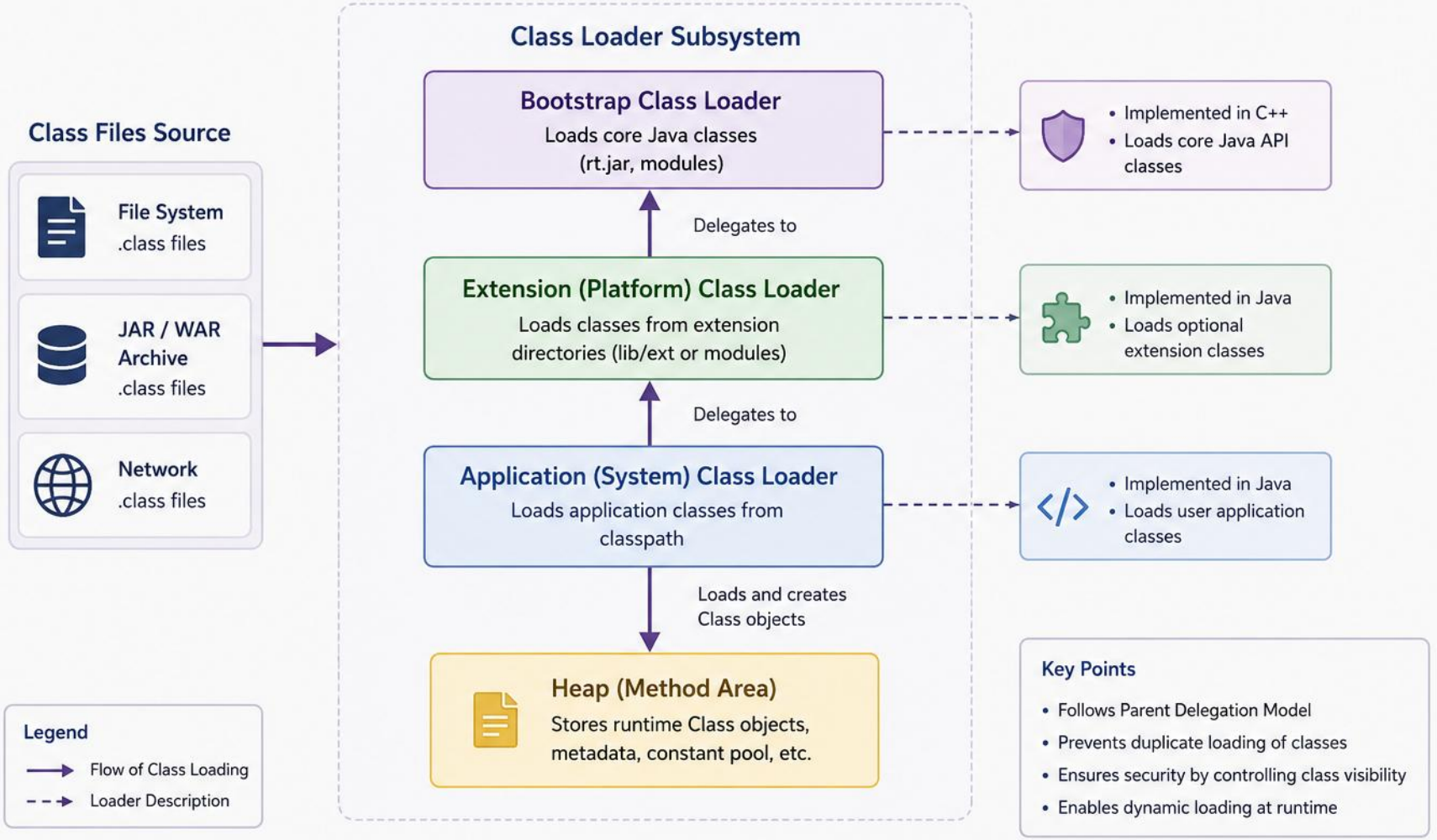
- Loads from the application classpath (-cp).
- e.g. com.myapp.*, com.example.*

CLASS LOADING LIFECYCLE Every class passes through three phases — Loading, Linking (verify, prepare, resolve), Initialization (static vars/blocks run) — via the 6-step flow: Request → Delegate → Load → Link → Initialize → Ready.

KEY TAKEAWAY Requests delegate upward through the hierarchy — this is the parent delegation model, and it exists for security.

Class Loader Subsystem

Dynamically loads .class files into the JVM and creates Class objects in the Heap.



RUNTIME DATA AREAS

The JVM manages memory using runtime data areas, each with a specific purpose and lifecycle.

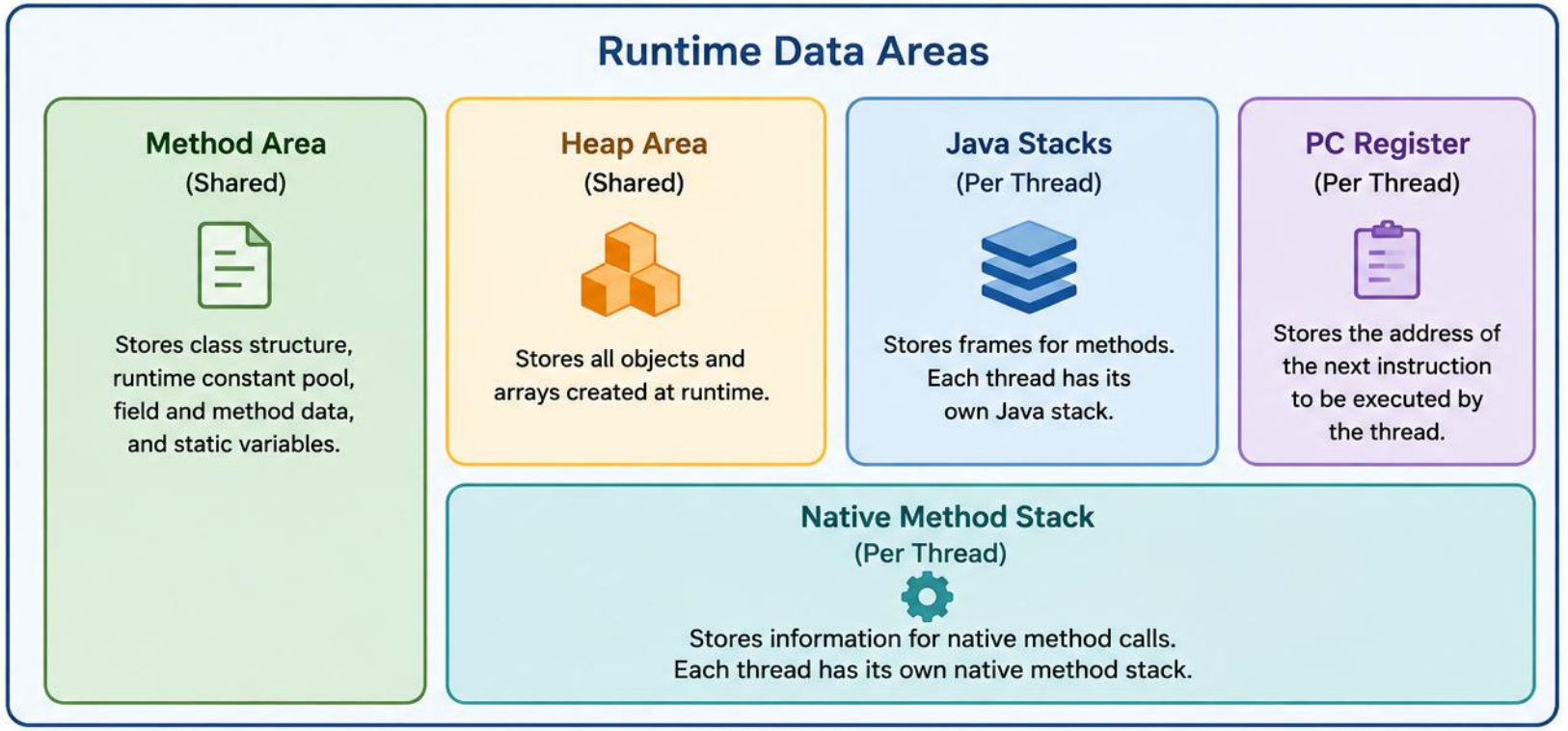
AREA	SCOPE	PURPOSE
`Method Area`	Shared	Class-level data: constant pool, field/method data, static variables.
`Heap Area`	Shared	All objects and arrays; managed by the Garbage Collector; the largest area.
`Java Stacks`	Per-thread	Frames per method invocation: local variables, operand stack, constant-pool reference.
`PC Register`	Per-thread	Address of the next instruction to execute; undefined for native methods.
`Native Method Stack`	Per-thread	Supports native (C/C++) method calls, created and destroyed with the thread.
`Runtime Constant Pool`	Part of Method Area	Constants used in compiled code: literals and symbolic references.

WHY THIS MATTERS Understanding runtime areas drives memory tuning, performance optimization, diagnosing OutOfMemoryError, JVM monitoring, and building robust, scalable applications.

KEY TAKEAWAY Memory is allocated and reclaimed automatically — but OutOfMemoryError can still occur if memory runs out.

JVM Runtime Data Areas

Memory areas used by the JVM during program execution



Method Area
Shared among all threads. Created at JVM start-up and exists until JVM exits.

Heap Area
Shared among all threads. Objects are allocated here using new keyword.

Java Stacks
Created when a thread starts. Destroyed when the thread terminates.

PC Register
Created for each thread. Stores address of next instruction.

Native Method Stack
Used for native (JNI) method calls. Created per thread.

All threads share the Method Area and Heap Area.
Each thread has its own Java Stack, PC Register, and Native Method Stack.

EXECUTION ENGINE

The heart of the JVM — it executes bytecode, optimizes performance, and works with Runtime Data Areas and native libraries.

KEY RESPONSIBILITIES

- Execute bytecode: interprets and executes instructions.
- Optimize performance using JIT compilation and other techniques.
- Manage runtime operations: object creation, synchronization, exception handling, method invocation.
- Interface with native libraries via JNI when needed.

COMPONENTS OF THE EXECUTION ENGINE

Interpreter

- Reads bytecode instruction by instruction; fast startup.

JIT Compiler

- Compiles hot spots to native code for throughput.

GC Interface

- Works with the Garbage Collector to reclaim objects.

Runtime Support

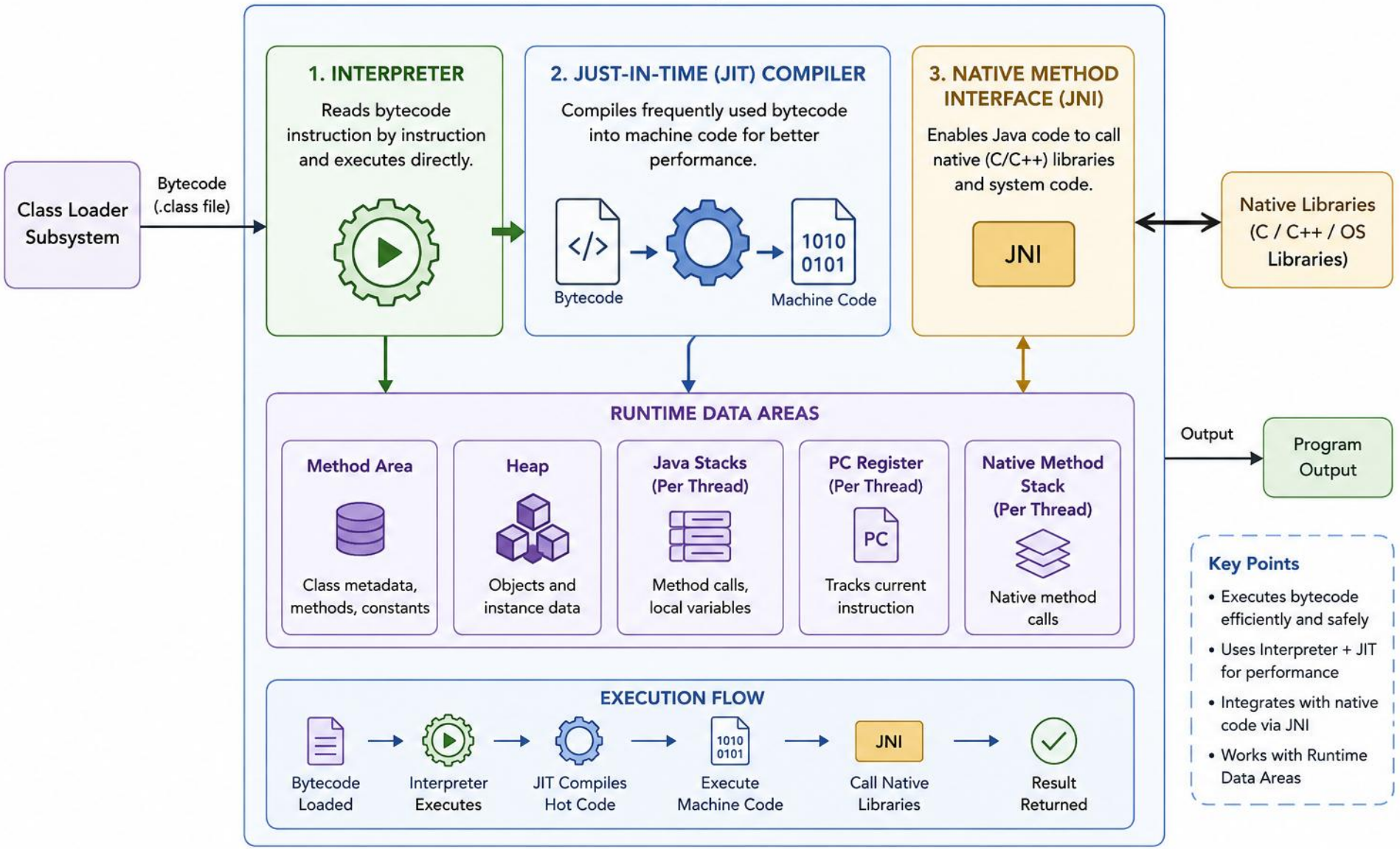
- Threading, sync/locking, exceptions, deoptimization.

EXECUTION FLOW Bytecode Loaded → Fetch Instructions → Interpret/JIT → JIT Compiles Hot Code → Runtime + GC → Result. JIT detects hot spots and compiles them to native code, which is why Java gets faster after warm-up.

KEY TAKEAWAY The Execution Engine works with memory areas, the class loader, and the garbage collector for correct, efficient execution.

EXECUTION ENGINE

Executes bytecode and manages runtime operations



JAVA NATIVE INTERFACE (JNI)

A standard mechanism that lets Java code in the JVM call, and be called by, native applications written in C or C++.

WHY JNI?

Reuse Native Code

Leverage existing libraries and legacy code.

Performance

Performance-critical operations in native code.

Platform Features

Access platform-specific capabilities.

Integration

Connect with non-Java systems and frameworks.

HOW JNI WORKS

1. **Calls Native Method**
2. **JVM Loads Library**
3. **JNI Bridges**
4. **Native Code Runs**
5. **Result to Java**

KEY TAKEAWAY JNI does not break "Write Once, Run Anywhere" — it loads the appropriate native library for each platform.

JNI — DATA TYPES & EXAMPLE

Every Java type maps to a specific C/C++ JNI type.

JAVA TYPE	JNI TYPE	JAVA TYPE	JNI TYPE
boolean	jboolean	**char**	jchar (unsigned short)
byte	jbyte	**short**	jshort
int	jint	**long**	jlong
float	jfloat	**double**	jdouble
Object	jobject	**String**	jstring
void	void	**Array**	j<type>Array

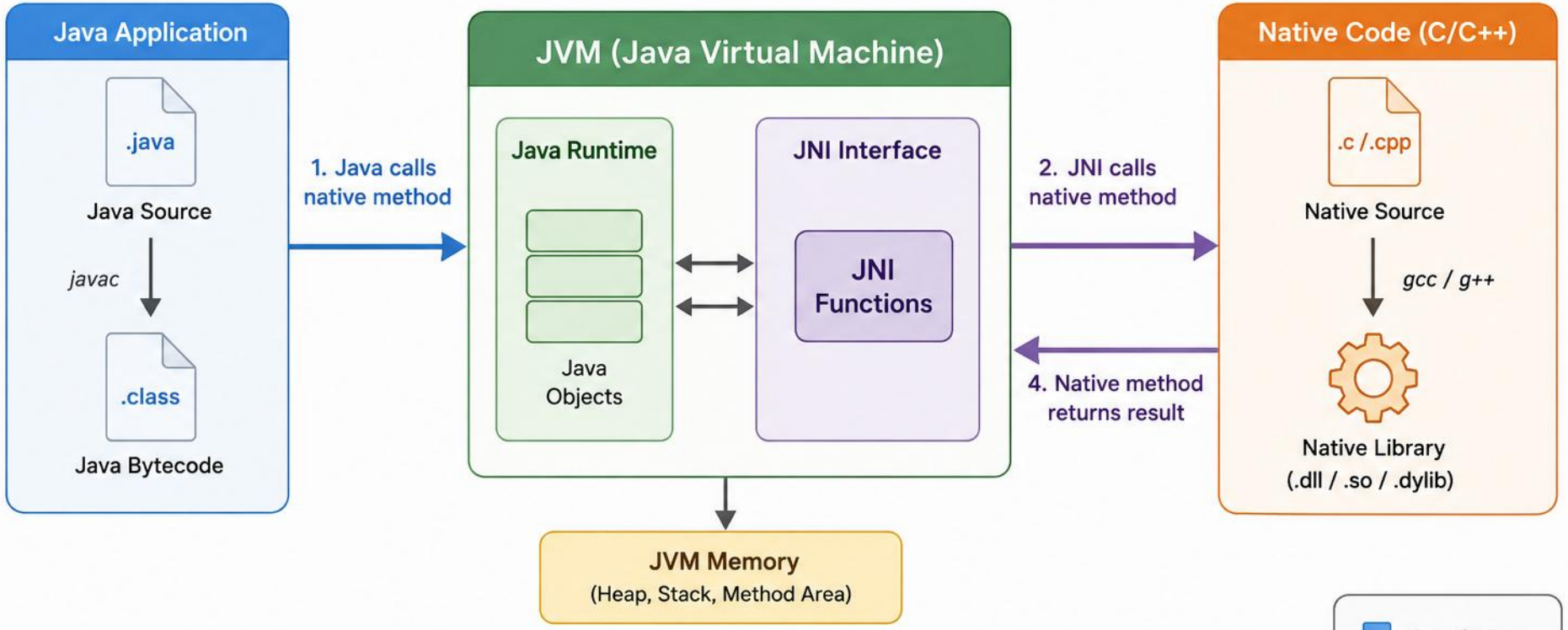
```
public class MyClass {
  native void nativeMethod();
  static { System.loadLibrary("native"); }
}
```

```
#include <jni.h>
JNIEXPORT void JNICALL
Java_MyClass_nativeMethod(JNIEnv *e, jobject o) {
  // native C implementation
}
```

KEY TAKEAWAY Always release native resources, follow naming conventions for native methods, handle exceptions across the boundary, and test on every target platform.

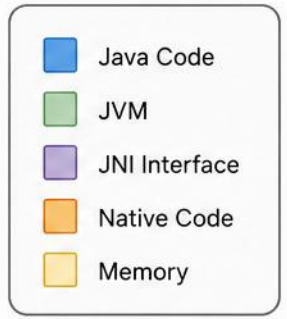
Java Native Interface (JNI)

JNI enables Java code running in the JVM to call native code written in languages like C or C++ and vice versa.



How JNI Works

1. Java code calls a native method.
   `native void myMethod();`
2. JVM uses JNI to locate and call the native method.
3. Native method executes (C/C++ code).
4. Control returns to JVM and the result is returned to Java code.



NATIVE METHOD LIBRARIES

Collections of native (non-Java) code that Java programs call through JNI, extending Java with platform-specific capability.

KEY FEATURES

Platform Specific

OS/hardware features beyond standard APIs.

Performance

High-performance optimized native code.

Extensibility

Integrate legacy/native libraries.

Reliability

Well-tested native libraries improve stability.

Seamless Integration

Invoked as native methods via JNI.

HOW NATIVE LIBRARIES WORK

1. **Calls Native Method**
2. **JVM Loads Library**
3. **JNI Bridges**
4. **Native Code Runs**
5. **Result to Java**

REAL-WORLD EXAMPLES OpenSSL (crypto/SSL, all platforms) • Graphics Libraries – OpenGL/DirectX (games, 3D, platform-specific) • Multimedia Libraries (audio/video, all platforms) • System Libraries – zlib/ICU (compression, i18n, all platforms).

KEY TAKEAWAY Native libraries are platform-dependent: DLL on Windows, .so on Linux, .dylib on macOS — matched to 32-bit/64-bit architecture.

COMMON NATIVE METHOD LIBRARIES IN PRACTICE

Real-world Java APIs that are backed by native libraries under the hood.

OpenGL

2D/3D graphics rendering.

Oracle/MySQL JDBC Drivers

Native database connectivity.

Java Sound (PortMixer/ALSA/CoreAudio)

Audio I/O.

Java Advanced Imaging (JAI)

Image processing.

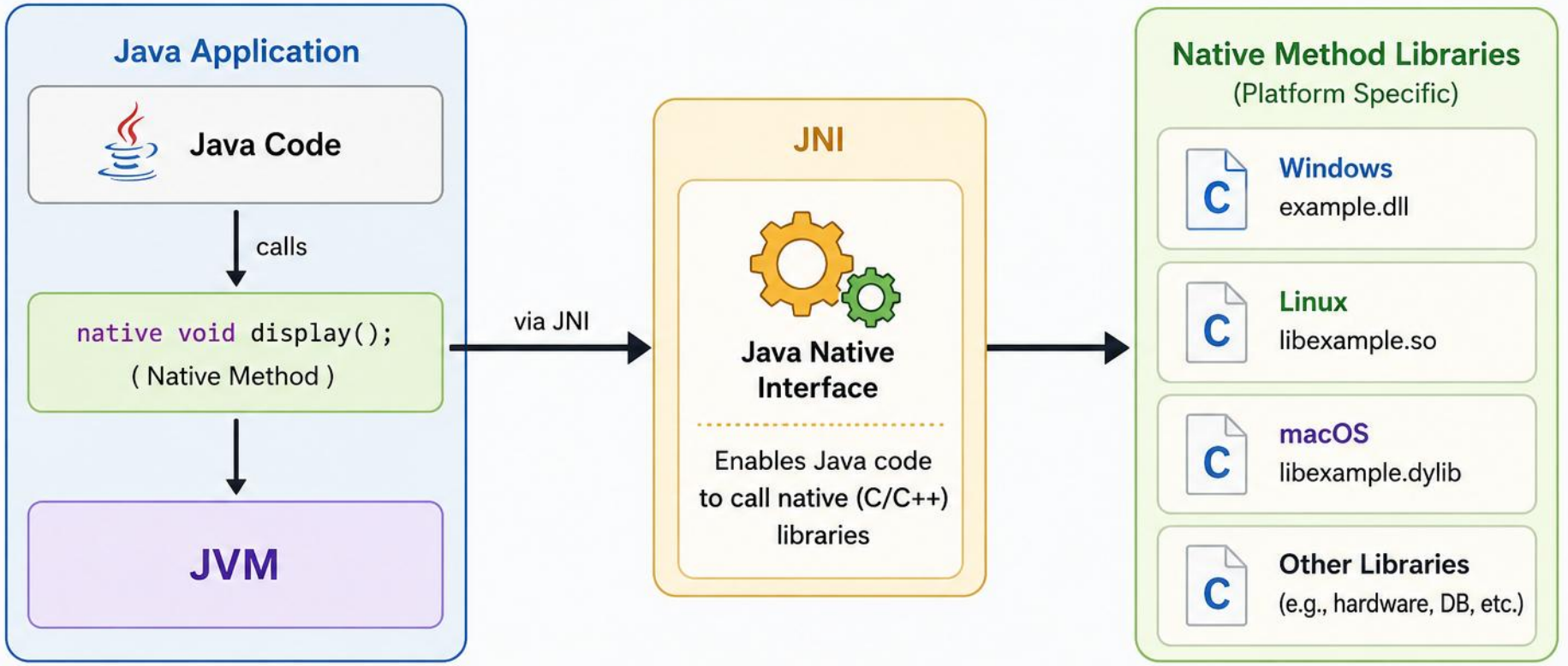
Netty/OpenSSL

Networking and SSL.

KEY TAKEAWAY Many everyday Java APIs — graphics, databases, audio, networking — are Java-friendly wrappers around native libraries.

Native Method Libraries

Native method libraries contain platform-specific code that can be invoked by Java applications through the JNI.



- Native method libraries are written in languages like C/C++.
- They provide access to platform-specific features and optimized performance.
- Examples: File I/O, Network operations, OS APIs, Hardware integration.

INSIDE THE JVM — PUTTING IT TOGETHER

The four subsystems that cooperate on every run.

Class Loader Subsystem

- Loading → Linking → Initialization; delegation flows Bootstrap → Platform → Application.

Runtime Data Areas

- Method Area, Heap, Java Stacks, PC Register, Native Method Stack.

Execution Engine

- Interpreter executes bytecode; JIT compiles hot code; GC reclaims memory.

Native Integration via JNI

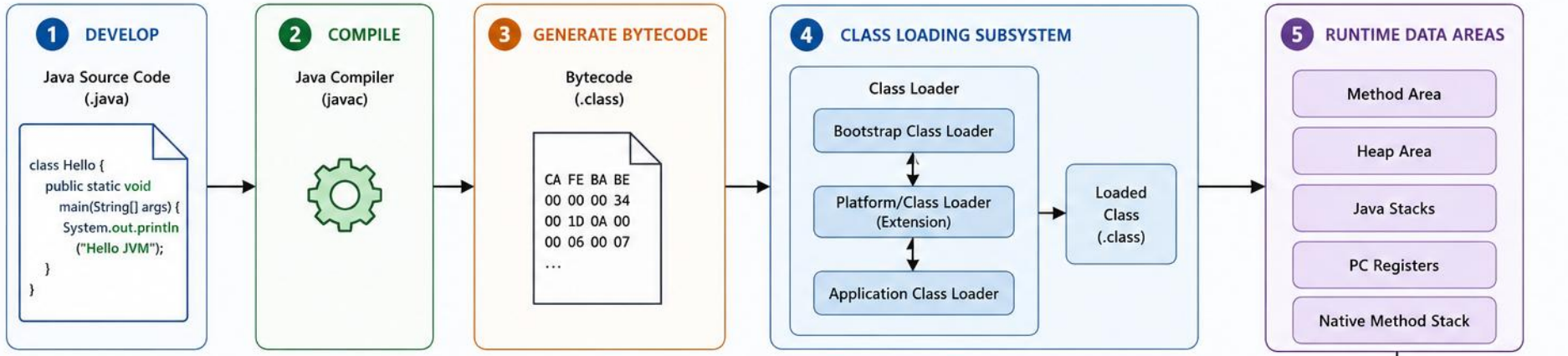
- Java Code ↔ JNI ↔ Native Libraries (.dll / .so / .dylib).

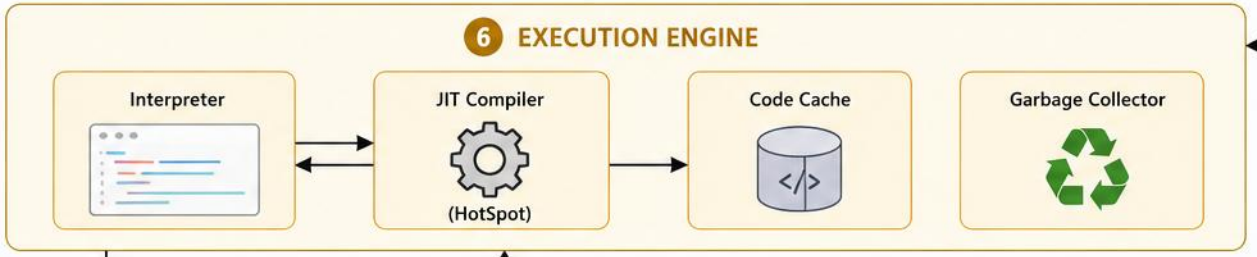
- Benefits of this architecture: platform independence, security, performance, scalability and extensibility, and reliability.

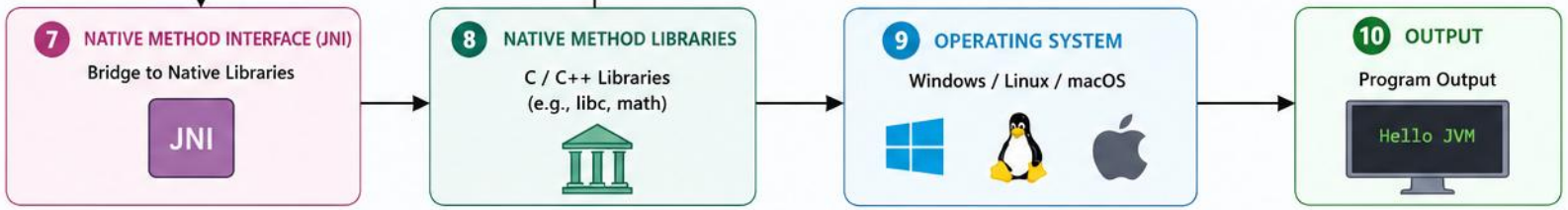
KEY TAKEAWAY JIT compilation improves performance over time, and the Garbage Collector automatically manages memory throughout.

END-TO-END JVM EXECUTION FLOW

From Java Source Code to Program Execution







FLOW SEQUENCE: 1 Develop → 2 Compile → 3 Bytecode → 4 Class Loading → 5 Runtime Data Areas → 6 Execution Engine → 7 JNI → 8 Native Libraries → 9 OS → 10 Output

JVM abstracts the underlying OS and hardware, providing a platform-independent runtime environment.

FROM SOURCE CODE TO BYTECODE

How Java code becomes platform-independent bytecode.

1. Write Source Code (.java file, text editor).
2. Invoke Compiler: javac from the command line.
3. Syntactic & Semantic Analysis: types, declarations, syntax.
4. Generate Bytecode: platform-independent .class file.
5. Ready for JVM: executable by any JVM, any platform.

```
$ javac HelloWorld.java
Compilation Successful. No errors found.
Bytecode generated.
```

FILE	GENERATED BY
HelloWorld.java	Developer
HelloWorld.class	javac Compiler

BYTECODE CHARACTERISTICS Java bytecode is stack-based (an operand stack, not CPU registers), strongly typed, and verified for safety by the JVM before it ever runs. Every .class file also carries metadata alongside the instructions — a magic number (0xCAFEBABE), constant pool, and access flags.

KEY TAKEAWAY The compiler does not create machine code — it creates bytecode that is executed by the JVM.

FROM SOURCE CODE TO BYTECODE

1. SOURCE CODE
(Hello.java)

```
public class Hello {
    public static void main
     (String[] args) {
        System.out.println
         ("Hello, World!");
    }
}
```

2. COMPILER
(javac)

The Java Compiler (javac) compiles the source code into bytecode.

3. BYTECODE
(Hello.class)

BYTECODE (JVM Instructions)

```
 0: aload_0
 1: invokespecial  #1
 4: getstatic      #2
 7: ldc            #3
 9: invokevirtual  #4
12: return
```

4. JVM EXECUTION

The JVM loads the bytecode from the .class file and executes it on any platform.

1. Source Code
Developer writes code in a .java file using any text editor or IDE.

2. Compiler (javac)
The Java compiler checks for syntax and semantic errors and converts the source code into bytecode.

3. Bytecode (.class)
Bytecode is platform-independent and stored in a .class file. It contains JVM instructions.

4. JVM Execution
The JVM loads the bytecode and executes it after just-in-time (JIT) compilation to native machine code.

Key Point: Bytecode acts as an intermediary between source code and machine code, enabling Java's platform independence (Write Once, Run Anywhere).

BYTECODE INSTRUCTIONS IN ACTION

Categories of instructions, and a stack trace for int c = a + b.

CATEGORY	PURPOSE	EXAMPLE INSTRUCTIONS
Load & Store	Move data between locals/stack	iload, aload, istore, astore
Arithmetic	Perform arithmetic operations	iadd, isub, imul, idiv
Object Operations	Create objects, access fields	new, getfield, putfield
Method Invocation	Call methods	invokestatic, invokevirtual
Control Flow	Branching and looping	ifeq, ifne, goto
Type Conversion	Convert between primitive types	i2l, i2d, i2b, i2c
Return	Return from a method	ireturn, areturn, return
Exception Handling	Throw and catch exceptions	athrow, try-catch blocks

```
int c = a + b;   // iload_1 (a) → iload_2 (b) → iadd → istore_3 (c)
```

KEY TAKEAWAY Bytecode is the heart of Java's portability and security — compact, efficient, and safe. Explore it with javap, ASM, or CFR.

Understanding Java Bytecode

Java bytecode is an intermediate representation of Java source code.
It is platform-independent, generated by the Java compiler (.class files),
and executed by the JVM on any platform.

1. Java Source Code

```
public class Hello {
    public static void
    main(String[] args) {
      System.out.println
      ("Hello, World!");
    }
}
```

2. Compiler (javac)

javac
converts source code into bytecode

3. Bytecode (.class file)

Hello.class

```
0:  aload_0
1:  invokespecial #1
4:  getstatic     #2
7:  ldc           #3
9:  invokevirtual #4
12: return
```

What is Bytecode?

- Low-level instructions
- Platform-independent
- Stored in .class files
- Not human readable

4. JVM Execution

JVM

JVM reads bytecode and executes it on the host machine

5. Output

```
Hello,
World!
```

Why Bytecode?

Platform Independence
Run once, run anywhere (WORA)

Security
Bytecode verification ensures safe execution

Optimization
JVM can optimize bytecode for better performance

Portability
Same .class files run on any system with a compatible JVM

End-to-End Flow

1. Write Java code (.java)
2. Compile with javac
3. Generates bytecode (.class)
4. JVM loads and verifies bytecode
5. JVM executes bytecode
6. Output is produced

Key Takeaway: Java bytecode is the bridge between your Java program and the platform it runs on, enabling portability, security, and performance.

TRY IT YOURSELF — EXERCISE 3: CONTROL FLOW

Reinforces: the branch and jump bytecode instructions you just saw.

STEP	WHAT TO DO
Goal	Create ControlFlow.java using if, for, while, and switch
File	ControlFlow.java (in examples/module-01-exercises/)
Compile	javac ControlFlow.java
Run	java ControlFlow — runs all four demos
Key concept	Each if/switch branch compiles to a conditional jump (e.g. ifeq, goto) like the ones you just saw
Reference	exercises/exercise-03-control-flow.md

```
$ cd examples/module-01-exercises
$ javac ControlFlow.java
$ java ControlFlow  // prints even/odd, a countdown, and a weekday from switch
$ javap -c ControlFlow  // look for ifeq / goto — the branch instructions behind if and while
```

TRY IT NOW Complete Exercise 3 before continuing — see exercises/exercise-03-control-flow.md.

JAVA COMPILATION PROCESS

From .java source code to .class bytecode — the stages javac performs.

COMMAND	DESCRIPTION
`javac Hello.java`	Compiles Hello.java to Hello.class
`javac *.java`	Compiles all Java files in current directory
`javac -d out Hello.java`	Places the .class file in the out directory
`javac -cp lib/* Hello.java`	Uses libraries in lib for compilation
`javac -verbose Hello.java`	Shows detailed compilation steps
`javap -c Hello.class`	Disassembles bytecode in human-readable form

COMPILER INTERNALS 5 phases: Lexical → Syntax → Semantic → IR → Bytecode Gen, via a symbol table.

KEY TAKEAWAY Compilation happens once; execution can happen many times — and every top-level class produces its own .class file.

Java Compilation Process

From Source Code to Bytecode

1. Write Source Code

.java

```
public class Hello {
   public static void
   main(String[] args) {
      System.out.println
      ("Hello, Java!");
   }
}
```

Hello.java

Developer writes Java source code in a .java file.

2. Compiler (javac)

javac

Java Compiler

The Java compiler (javac) checks the source code for syntax and semantic errors and converts it into bytecode.

3. Bytecode Output

.class

```
CA FE BA BE 00 00 00 34
00 0D 0A 00 03 00 09 07
00 0A 07 00 0B 01 00 10
6A 61 76 61 2F 6C 61 6E
67 2F 4F 62 6A 65 63 74
01 00 06 3C 69 6E 69 74
3E 01 00 03 28 29 56 ...
```

Hello.class

The output is platform-independent bytecode stored in a .class file.

4. Ready for Execution

Java

JVM

The .class file is loaded by the JVM and executed. JVM converts bytecode into machine code at runtime.

Key Points

- javac compiles .java files into .class files.
- Bytecode is platform-independent and runs on any system with a JVM.
- Compilation happens once; execution can happen many times.
- Errors in source code are caught at compile time.

INTRODUCTION TO CLASS LOADING

How the JVM dynamically locates, loads, and prepares classes for execution.

- When a class is needed, the JVM locates the .class file, loads the class data, links it (verify/prepare/resolve), and initializes it (runs static initializers). After this, the class is ready to use.

CLASS LOADER	PARENT	RESPONSIBILITY	LOADS FROM
`Bootstrap`	None (root)	Loads core Java classes	rt.jar / Java 9+ modules
`Platform (Extension)`	Bootstrap	Loads extension libraries	jre/lib/ext or extensions
`Application (System)`	Platform	Loads application classes	CLASSPATH, -cp, jars
`Custom`	Application	Loads classes per app needs	Network, DB, user-defined

KEY TAKEAWAY Parent Delegation Model: a class loader first delegates to its parent; if the parent can't find the class, it tries to load it itself.

WHEN IS A CLASS LOADED?

Not every reference to a class causes it to be loaded — only active use does.

ROLE OF THE CLASS LOADER SUBSYSTEM

A CLASS IS LOADED WHEN:

- It is first actively used: new, a static method call, or accessing a static variable.
- java.lang.Class.forName("...") is called.
- A subclass is used, or an interface with default methods is used.

WHY CLASS LOADING MATTERS

Security

Prevents unauthorized classes.

Efficiency

Loads classes on demand.

Flexibility

Supports custom loaders.

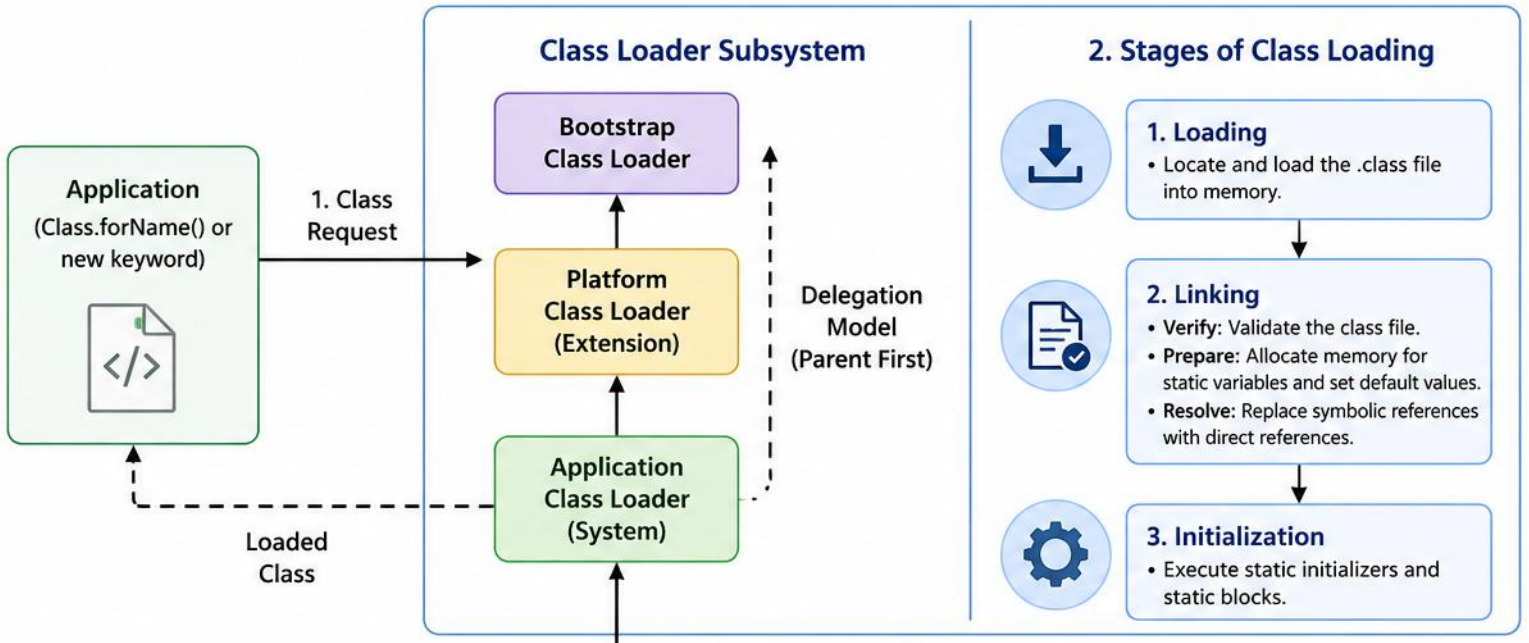
Platform Independence

Enables portability.

KEY TAKEAWAY Understanding class loading helps in debugging ClassNotFoundException and NoClassDefFoundError, and in building frameworks, containers, and custom class loaders.

Java Class Loading

The Class Loader Subsystem is responsible for loading .class files into the JVM.

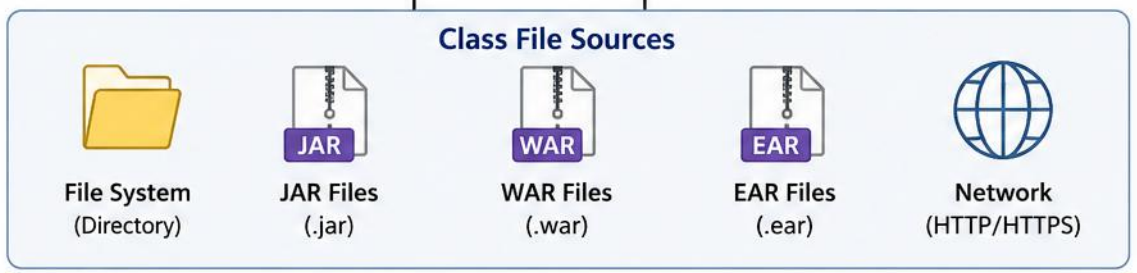


2. Stages of Class Loading

1. Loading
- Locate and load the .class file into memory.

2. Linking
- **Verify:** Validate the class file.
- **Prepare:** Allocate memory for static variables and set default values.
- **Resolve:** Replace symbolic references with direct references.

3. Initialization
- Execute static initializers and static blocks.



Key Points
- Uses Delegation Model (Parent First) for security and stability.
- Bootstrap loads core Java classes (rt.jar / modules).
- Application Class Loader loads application-specific classes.

Benefits
- **Security:** Prevents overriding core classes.
- **Modularity:** Different class loaders can load different versions of classes.
- **Flexibility:** Supports dynamic loading at runtime.

- **Bootstrap Class Loader:** Loads core Java classes from JDK (e.g., rt.jar or modules).
- **Platform Class Loader:** Loads classes from JRE libraries (e.g., $JAVA_HOME/lib/ext or modules).
- **Application Class Loader:** Loads classes from classpath (e.g., user-defined classes).

BOOTSTRAP CLASS LOADER

The root of Java's class-loading hierarchy — the built-in native class loader in the JVM.

Native & Core

- Implemented in native code (C/C++); part of the JVM.

Always Trusted, No Parent

- Root of the hierarchy; classes cannot be overridden.

1 Loads Once

- Classes loaded once and cached in the JVM.

PACKAGE PREFIX	EXAMPLES OF CLASSES
`java.*`	java.lang.String, java.util.List, java.lang.Object, java.io.InputStream
`javax.*`	javax.crypto.Cipher, javax.xml.parsers.DocumentBuilder
`jdk.* (Java 9+)`	jdk.internal.misc.Unsafe, jdk.nio.*
`sun.*`	sun.misc.Unsafe, sun.reflect.Reflection

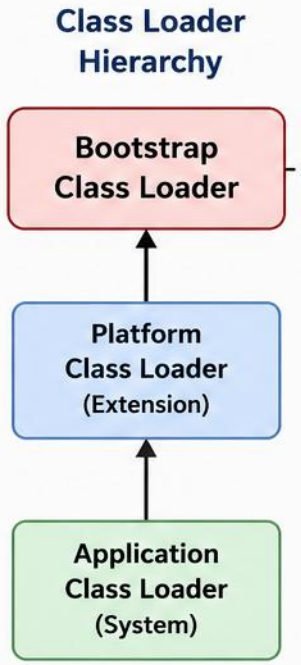
IN CODE String.class.getClassLoader() returns null — proof the class was loaded natively by Bootstrap, which has no Java-level representation. If Bootstrap can't find a class, it delegates down to Platform.

KEY TAKEAWAY You cannot directly extend or customize the Bootstrap Class Loader, and it never loads application or user-defined classes.

Bootstrap Class Loader

The root of Java Class Loading Hierarchy

Bootstrap Class Loader is the first and highest priority class loader.
It loads core Java classes from the JDK and provides the foundational classes required by the JVM.



Bootstrap Class Loader

What it does

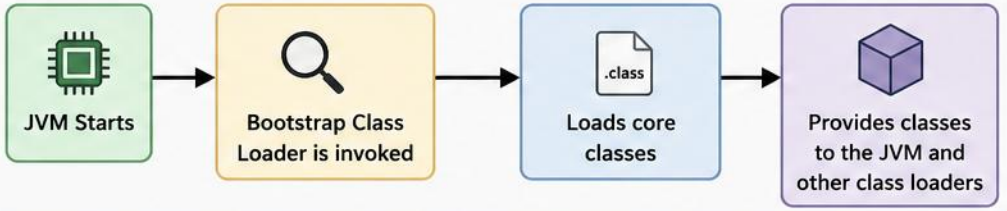
- Loads core Java classes from the JDK
- Implemented in native code (C/C++)
- No parent (it is the root class loader)

What it loads

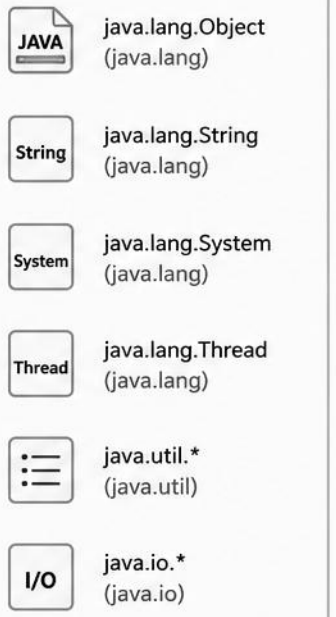
Loads classes from the $JAVA_HOME/jre/lib (or $JAVA_HOME/lib in modern JDKs)

- rt.jar (Java 8 and below)
- modules (Java 9 and above)

How it works



Examples of Classes Loaded



Key Points

- Highest priority class loader
- Implemented in native code
- Loads essential Java runtime classes
- Every other class loader depends on it

Note: The Bootstrap Class Loader is not a Java object and cannot be accessed directly via Java code.
It is implemented in the JVM and always present.

PLATFORM CLASS LOADER (EXTENSION CLASS LOADER)

The bridge between core Java and applications — child of Bootstrap, parent of Application.

- Implemented in Java, unlike Bootstrap which is native.
- Java 8 and earlier: loads from $JAVA_HOME/jre/lib/ext/*.jar (Extension Class Loader). Java 9+: loads from JDK modules instead.

WHAT DOES IT LOAD?

CATEGORY	EXAMPLES
`Platform Modules (Java 9+)`	java.sql, java.management, jdk.crypto.*
`Java Standard Extensions`	javax.*, org.w3c.dom, org.xml.sax
`User/Custom Platform Libraries`	libraries added via ext dir or --module-path

IN CODE Delegation goes up first: App → Platform → Bootstrap, proven by getClassLoader().

KEY TAKEAWAY The Platform Class Loader ensures both the JDK and applications have access to essential extension APIs.

Platform / Extension Class Loader (Java)

Loads platform specific classes and extension libraries that are not part of the core JDK but are provided to extend the Java SE platform.

Class Loader Hierarchy

Bootstrap Class Loader
Loads core Java classes (rt.jar, java.*)

Delegates (Parent-First)

Platform / Extension Class Loader
Loads extension classes (javax.*, org.w3c.* etc.)

Delegates (Parent-First)

Application Class Loader
Loads application classes (from classpath)

How Platform / Extension Class Loader Works

Platform / Extension Class Loader

1. Receives class loading request
2. Delegates to Bootstrap Class Loader
3. If not found, loads from extension directories
4. Defines the class and returns

Search Path

JDK
$JAVA_HOME/jre/lib/ext (or) $JAVA_HOME/lib/ext
(Legacy Extension Directory)

Extension Libraries
JAR JAR JAR
*.jar files placed in ext directory

Automatic Modules
Automatic modules from jars on the extension path

Loaded Classes
Defined and made available to the application

Key Points

- Sits between Bootstrap and Application Class Loader.
- Loads libraries that extend Java SE platform.
- Follows Parent-First delegation model.
- In modern Java (9+), extension mechanism is replaced by the Module System.

Note: The traditional extension mechanism (ext directory) is deprecated in Java 9 and removed in Java 11. Use the module path and custom modules to extend the platform.

APPLICATION CLASS LOADER (SYSTEM CLASS LOADER)

The gateway for your application classes — the last loader in the delegation hierarchy.

- Child of the Platform Class Loader; implemented in Java; default loader for user-defined classes.
- Loads from the application classpath: current directory, JAR files, directories, -cp / -classpath.
- If Bootstrap and Platform loaders don't find a class, the request finally reaches this loader — if still not found, a ClassNotFoundException is thrown.

WHAT DOES IT LOAD?

TYPE	EXAMPLES
`User-defined Classes`	com.example.MyClass, org.myapp.service.UserService
`Application Libraries`	third-party JARs, custom utility libraries
`Application Packages`	com.company.app.*, org.project.module.*
`Resources (via ClassLoader)`	config.properties, log4j.xml, templates/

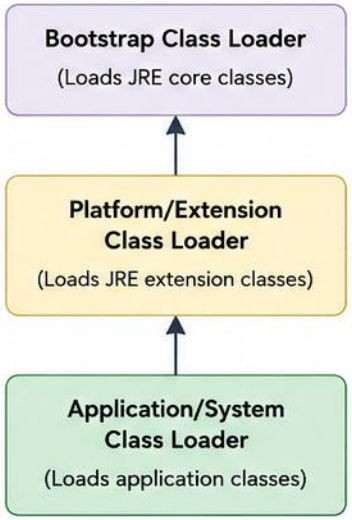
IN CODE getClassLoader() on a user class prints AppClassLoader@18b4aac2 — proof this loader (the one -cp/-classpath configures) handled it, only after Platform and Bootstrap both passed.

KEY TAKEAWAY Each class loader has its own namespace, enabling isolation — you can create custom loaders by extending ClassLoader.

Application/System Class Loader

The Application (or System) Class Loader is the default class loader that comes with the JVM. It is responsible for loading classes from the application classpath.

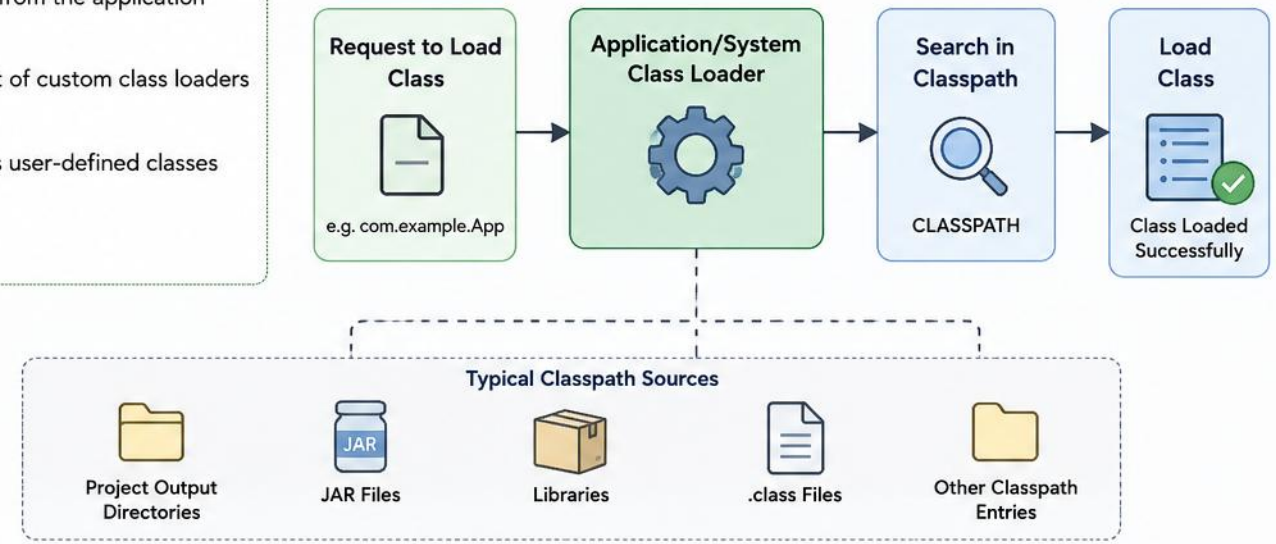
Class Loader Delegation Hierarchy



Role

- Loads classes from the application classpath.
- It is the parent of custom class loaders (if any).
- Typically loads user-defined classes and libraries.

Class Loading Flow



Key Points

- It is also called the "System Class Loader".
- It uses the parent delegation model.
- You can get a reference to it in Java using: `ClassLoader.getSystemClassLoader()`
- It loads classes from the `-cp` (or `-classpath`) specified when running the application.

Example (Java)

```
ClassLoader loader =
    ClassLoader.getSystemClassLoader();
System.out.println(loader);
// Output: sun.misc.Launcher$AppClassLoader@18b4aac2
```

CLASS LOADING LIFECYCLE

From .class file to ready-to-use class — seven phases, grouped into three stages.

PHASE	WHAT HAPPENS	KEY POINT
`Loading`	JVM locates .class file, reads binary, creates Class object.	Uses class loader hierarchy.
`Verification`	Verifies bytecode format & semantic correctness.	Protects JVM from invalid bytecode.
`Preparation`	Allocates memory for static fields, sets defaults.	No actual values assigned yet.
`Resolution`	Converts symbolic references to direct references.	May trigger loading of other classes.
`Initialization`	Executes static initializers (<clinit>), only once.	Only phase that runs app code.
`Using`	Class is ready for use.	Normal program execution phase.
`Unloading`	Class becomes eligible for GC.	Occurs when its ClassLoader is GC'd.

KEY TAKEAWAY A LinkageError is thrown if verification fails; classes loaded by different ClassLoaders are treated as different, even from the same .class file.

BENEFITS OF THE CLASS LOADING LIFECYCLE

Why the JVM enforces this exact seven-phase sequence, every time.

JVM Security & Stability

Ensures the runtime stays secure and stable.

Prevents Malicious Bytecode

Blocks execution of malicious or malformed bytecode.

Optimizes Memory

Unloading frees memory when classes are no longer needed.

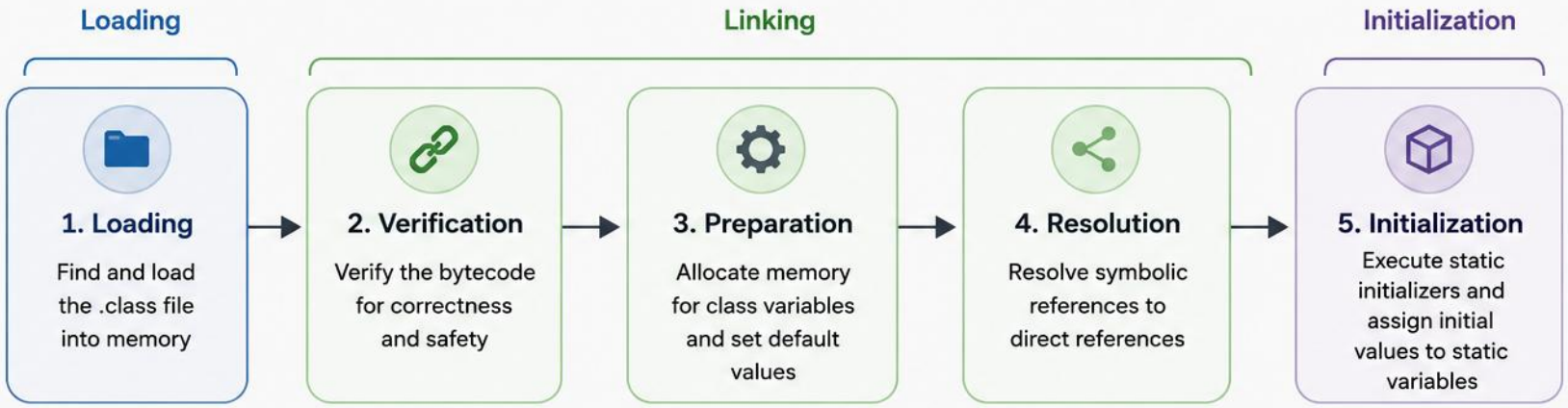
Structured Management

Gives the JVM a structured way to manage classes.

KEY TAKEAWAY If initialization fails, ExceptionInInitializerError is thrown — and unloading is never guaranteed, since it depends on garbage collection.

Class Loading Lifecycle

The process of loading a class into the JVM



Detailed Overview

1. Loading	The class loader locates the .class file (from classpath, JAR, etc.) and loads the binary data into the method area.
2. Verification	The JVM verifies the bytecode for format, semantic correctness, and security (type safety).
3. Preparation	Memory is allocated for the class's static variables and set to default values (0, null, false, etc.).
4. Resolution	Symbolic references (e.g., to classes, methods, fields) are resolved to direct references.
5. Initialization	Static initializers and static blocks are executed in order. Static variables are assigned their defined values.

Key Points

- Loading is done by Class Loader Subsystem.
- Linking is the process of preparing the class for use.
- Initialization occurs only once per class in a ClassLoader.
- After initialization, the class is ready for use.

TRY IT YOURSELF — EXERCISE 4: WATCH CLASS LOADING

Reinforces: the Bootstrap / Platform / Application class loaders you just saw.

STEP	WHAT TO DO
Goal	Run Hello with -verbose:class and see which loader loaded it vs. a JDK class
File	Hello.class (already compiled in Exercise 1)
Command	java -verbose:class Hello
Compare	java.lang.String is Bootstrap-loaded; Hello is Application-loaded
Key concept	Bootstrap loads core JDK classes; the Application loader loads your own classpath
Reference	exercises/exercise-04-class-loading.md

```
$ cd examples/module-01-exercises
$ java -verbose:class Hello
$ java -verbose:class Hello | findstr "Hello String"  // Windows: filter to Hello + String
$ java -verbose:class Hello | grep -E "Hello|String"  // macOS/Linux: same filter
```

TRY IT NOW Complete Exercise 4 before continuing — see exercises/exercise-04-class-loading.md.

JVM MEMORY ARCHITECTURE OVERVIEW

How the JVM organizes and manages memory — divided into specific runtime data areas.

SHARED BY ALL THREADS

- Method Area (Metaspace): class metadata.
- Runtime Constant Pool: literals & symbolic refs.
- Heap: all objects and arrays.
- Code Cache: JIT-compiled native code.

PRIVATE TO EACH THREAD

- JVM Stack: frames for method calls.
- Native Method Stack: native (C/C++) calls.
- PC Register: address of current instruction.

- The Method Area was called Permanent Generation (PermGen) in Java 7 and earlier; in Java 8+ it is Metaspace, outside the heap.
- The Heap is the only area where objects are allocated; use -Xms and -Xmx to set minimum/maximum heap size.

KEY TAKEAWAY StackOverflowError comes from exhausted stack memory; OutOfMemoryError comes from exhausted heap or metaspace.

JVM MEMORY — KEY FEATURES & TUNING

Automatic management, backed by a handful of tuning flags you'll use constantly.

Automatic Memory Mgmt

JVM allocates/reclaims automatically.

Platform Independence

Same program runs anywhere with a JVM.

Security

Isolated memory areas for safe execution.

Multithreading Support

Per-thread structures ensure safety.

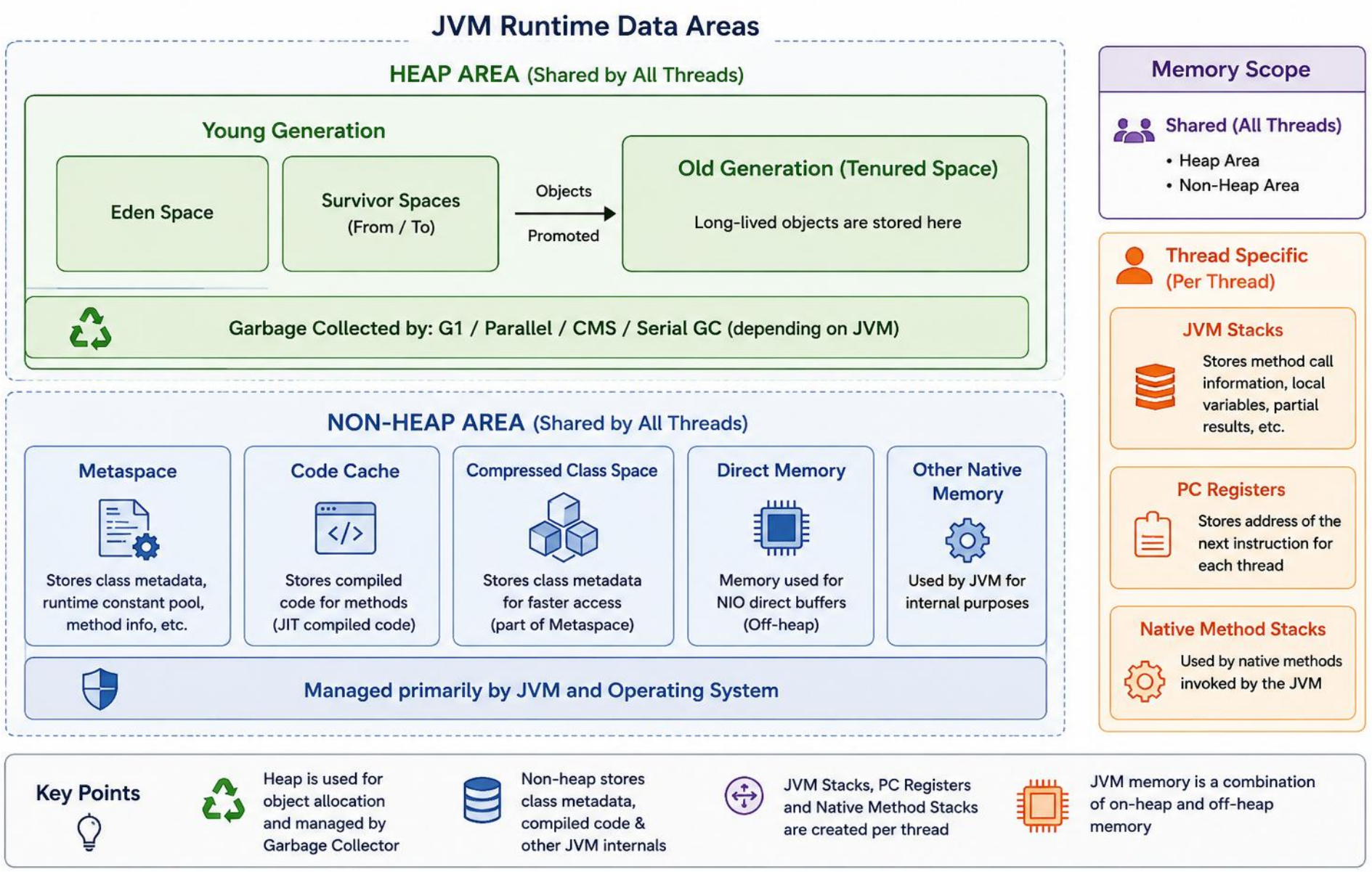
High Performance

JIT compilation and optimized memory use.

OPTION	PURPOSE
`-Xms<size> / -Xmx<size>`	Initial / maximum heap size
`-XX:MetaspaceSize / -XX:MaxMetaspaceSize`	Initial / maximum metaspace size
`-Xss<size>`	Thread stack size
`-XX:+UseG1GC`	Use the G1 Garbage Collector

KEY TAKEAWAY Monitor memory with JConsole, VisualVM, or JMC — and tune heap size based on real application needs, not guesswork.

JVM MEMORY ARCHITECTURE



HEAP MEMORY IN JAVA

The home of objects — a runtime data area shared by all threads and managed by the Garbage Collector.

- All object instances and arrays live here; created using the new keyword.
- Heap size is configurable; objects are reclaimed by the GC once no longer reachable.

HEAP STRUCTURE — GENERATIONS

Young Generation

- New objects in Eden; survivors move to S0/S1 after Minor GC.

Old Generation

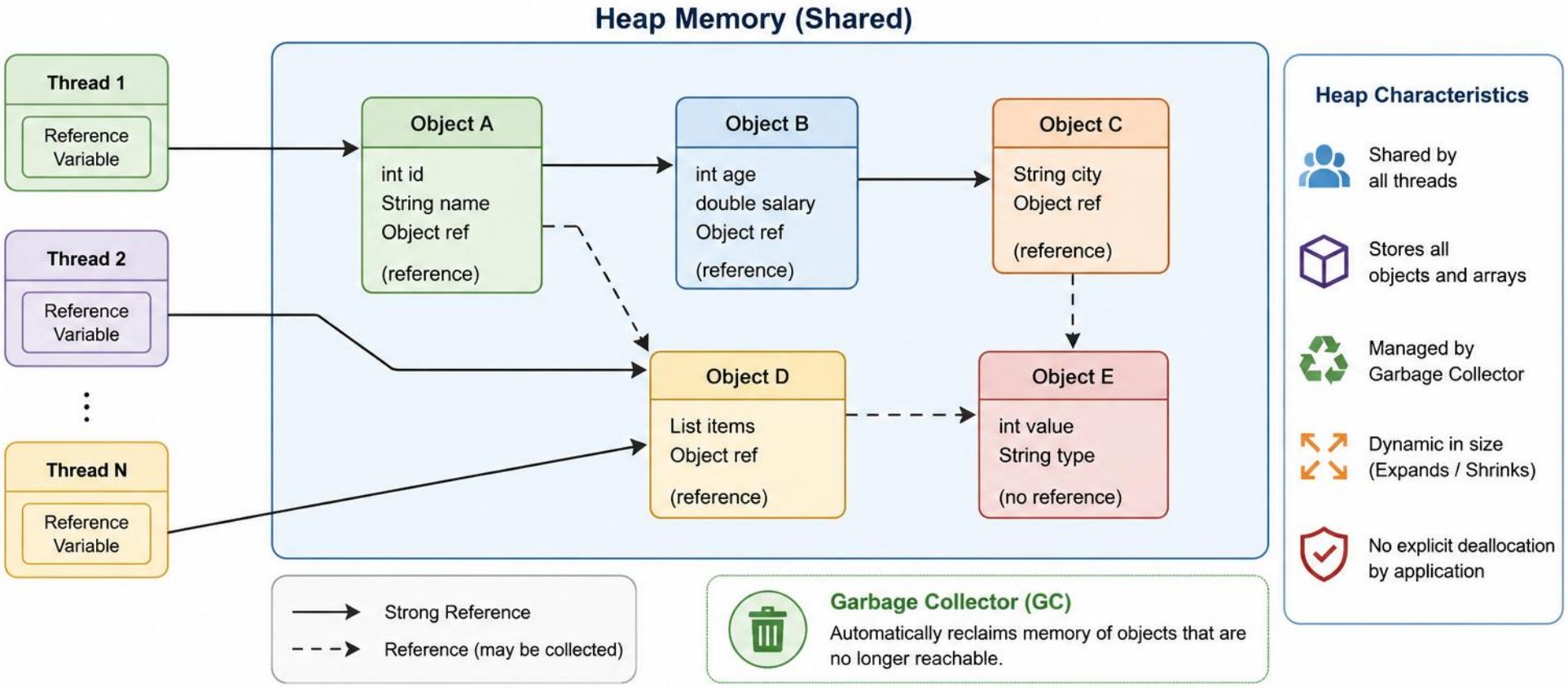
- Long-lived objects move here after multiple GC cycles; Major GC applies.

MINOR VS MAJOR GC Minor GC clears the Young Generation (short pause); Major/Full GC clears the Old Generation (longer pause) via mark-sweep from GC roots. Tune with -Xms/-Xmx (heap size) and -XX:+HeapDumpOnOutOfMemoryError to capture a snapshot on failure.

KEY TAKEAWAY Reference variables live in the Stack; the objects they point to live in the Heap — multiple references can share one object.

Java Heap Memory

The Heap is a runtime data area in the JVM where objects are allocated and stored.
It is shared by all threads and managed by the Garbage Collector.



Key Points

- Objects are created using 'new' keyword.
- All class instances and arrays are stored in the Heap.
- Objects become eligible for GC when no references point to them.
- Proper memory management improves application performance.

STACK MEMORY IN JAVA

The home of method calls and local variables — every thread has its own Stack.

- Stores local variables, method parameters, and return addresses; follows LIFO order.
- Memory is automatically allocated and deallocated — no manual management needed.
- Stack size is limited and typically smaller than the Heap.

STACK FRAME (ACTIVATION RECORD)

COMMON JVM OPTIONS FOR STACK

OPTION	PURPOSE
`-Xss<size>`	Sets thread stack size
`-XX:ThreadStackSize=<k>`	Sets thread stack size in KB
`-XX:+PrintFlagsFinal`	View the default stack size

KEY TAKEAWAY Default stack size varies by platform (typically 1MB on 64-bit systems) — the frame is pushed on call, popped on return.

STACK VS HEAP & COMMON ERRORS

The two memory areas every Java developer must be able to reason about.

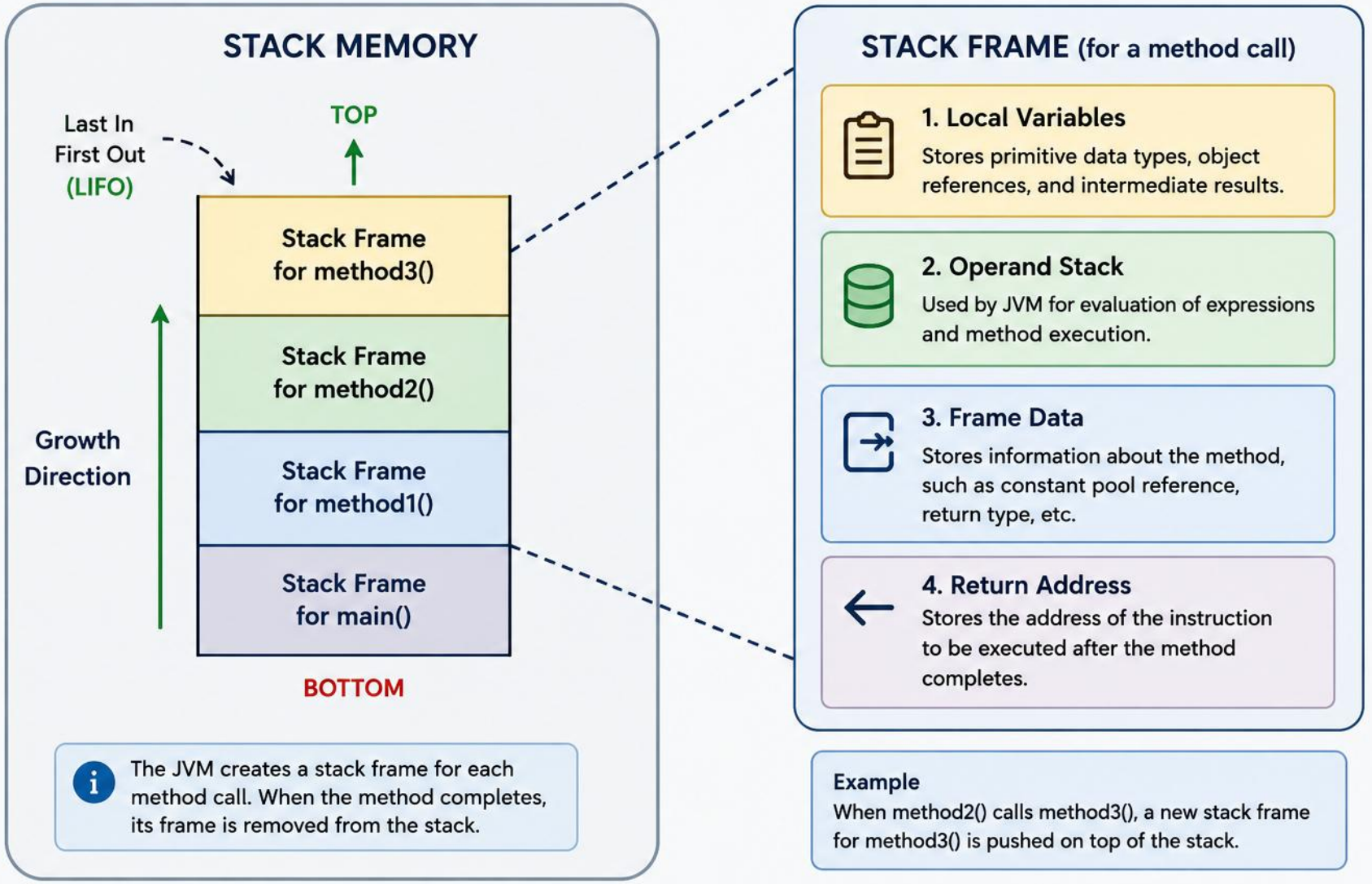
FEATURE	STACK	HEAP
`Scope`	Per thread	Shared by all threads
`Stores`	Locals, parameters, return address	Objects and arrays
`Size`	Smaller, limited	Larger, configurable
`Management`	Automatic push/pop	Garbage collected
`Access Speed`	Faster	Relatively slower
`Lifetime`	Method call ends	Until no references remain

- StackOverflowError: stack memory full due to deep recursion or too many nested calls.
- OutOfMemoryError: unable to create new native thread — insufficient stack memory or OS thread limits.

KEY TAKEAWAY Stack memory is fast, thread-specific, and automatically managed — critical for every method call in Java.

STACK MEMORY IN JAVA

The stack memory stores method calls and local variables.



TRY IT YOURSELF — EXERCISE 5: VARIABLES AND DATA TYPES

Reinforces: local variables live in the stack frame you just saw.

STEP	WHAT TO DO
Goal	Create Variables.java with several primitive types and one String; print each
File	Variables.java (in examples/module-01-exercises/)
Compile	javac Variables.java
Run	java Variables
Key concept	int, long, double, boolean, and char are stored directly in the method's local variable slots on the stack
Reference	exercises/exercise-05-variables.md

```
$ cd examples/module-01-exercises
$ javac Variables.java
$ java Variables  // prints age, population, price, enrolled, grade, and name
$ javap -c Variables  // istore / iload instructions show each local variable's stack slot
```

TRY IT NOW Complete Exercise 5 before continuing — see exercises/exercise-05-variables.md.

TRY IT YOURSELF — EXERCISE 6: METHODS AND PARAMETERS

Reinforces: every method call pushes a new frame onto the stack you just saw.

STEP	WHAT TO DO
Goal	Create Methods.java with two methods that take parameters and return a value
File	Methods.java (in examples/module-01-exercises/)
Compile	javac Methods.java
Run	java Methods — main calls add(...) and greet(...)
Key concept	Each call gets its own short-lived stack frame holding locals, parameters, and the return address
Reference	exercises/exercise-06-methods.md

```
$ cd examples/module-01-exercises
$ javac Methods.java
$ java Methods  // prints 30 and "Hello, Aman!"
$ javap -c Methods  // invokestatic calls add/greet; a new frame opens for each
```

TRY IT NOW Complete Exercise 6 before continuing — see exercises/exercise-06-methods.md.

METASPACE IN JAVA

The native memory for class metadata — replaces PermGen from Java 8 onward.

- Stores class metadata: class info, methods, fields, constant pool, and more.
- Allocated in native memory outside the JVM heap; grows dynamically as classes load.
- Class metadata is reclaimed when classes are unloaded and no longer in use.

FEATURE	PERMGEN (JAVA 7-)	METASPACE (JAVA 8+)
`Memory Location`	Inside JVM heap space	Native memory outside heap
`Size`	Fixed (-XX:PermSize/MaxPermSize)	Dynamic, grows as needed
`OutOfMemoryError`	OOME: PermGen space	OOME: Metaspace
`Tuning`	Difficult to tune	Easier, with simple limits
`Class Unloading`	Supported	Supported, with better scalability
`Default Size`	Smaller	Larger and more flexible

CLASS VS INSTANCE Metaspace stores information about a class (its blueprint) — never actual instances. Create 1,000 Person objects and Metaspace still holds exactly one Person blueprint; all 1,000 real objects live on the Heap.

KEY TAKEAWAY Metaspace is easier to tune than PermGen and scales better for class unloading.

METASPACE — FLOW & TUNING

Class loading feeds Metaspace; objects still go to the Heap.

OPTION	PURPOSE
`-XX:MetaspaceSize=<size>`	Initial metaspace size
`-XX:MaxMetaspaceSize=<size>`	Maximum metaspace size
`-XX:+PrintMetaspaceStatistics`	Print metaspace statistics
`-XX:+TraceClassLoading / Unloading`	Trace class loading/unloading
`-XX:MinMetaspaceFreeRatio=<n>`	Min free space % before GC (default 40)
`-XX:MaxMetaspaceFreeRatio=<n>`	Max free space % after GC (default 70)

INSPECT LIVE jcmd VM.metaspace shows used/committed/reserved; jstat -gc's M/CCS columns track it live.

KEY TAKEAWAY Metaspace grows dynamically, supports class unloading, and proper tuning helps avoid OutOfMemoryError: Metaspace.

OBJECT CREATION AND MEMORY ALLOCATION

From the new keyword to an object living in the Heap, with its reference in the Stack. Remember: == compares references, not contents; an object only becomes GC-eligible once unreachable (Creation → In Use → Unreachable → Reclaimed), never deleted directly.

#	STEP	DETAILS
1	new keyword	JVM encounters new Person(101, "Alice").
2	Class Loading	JVM loads Person if not already loaded.
3	Memory Allocation	Allocates Heap memory based on class structure.
4	Constructor Invocation	Calls Person(int, String) to initialize fields.
5	Reference Assignment	Reference stored in variable p in the stack frame.

KEY TAKEAWAY Objects are always created in the Heap; reference variables always live in the Stack.

MEMORY LAYOUT OF AN OBJECT IN HEAP

Every object on the Heap has the same three-part internal structure.

Object Header

- Class metadata, HashCode, GC info, and lock (monitor) info.

Instance Data

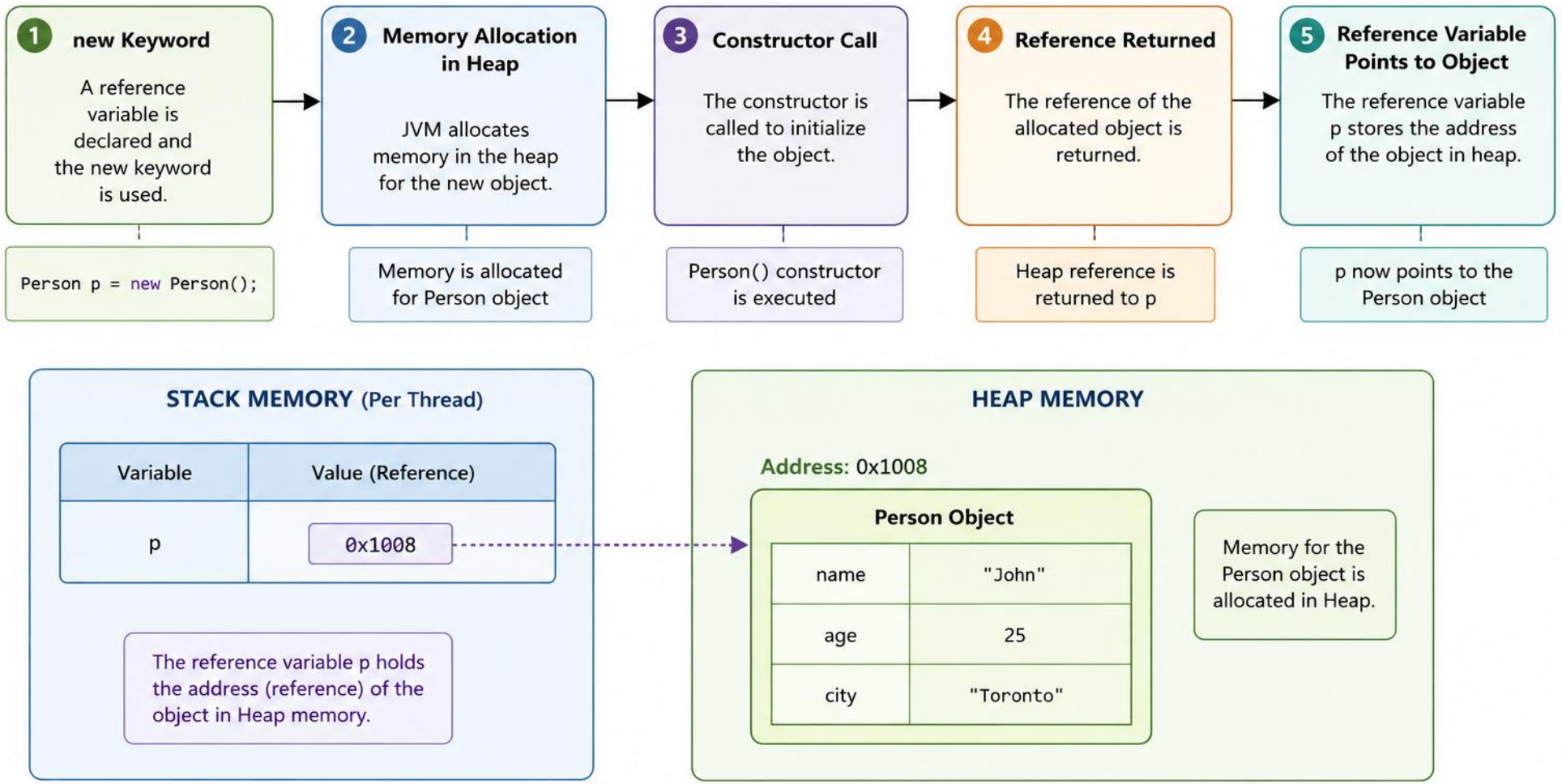
- The actual fields of the class and all its superclasses.

Padding / Alignment

- Optional padding so the object size aligns to 8-byte boundaries.

KEY TAKEAWAY Actual object size varies depending on the fields declared, the JVM implementation, and the CPU architecture.

Object Creation and Memory Allocation in Java



TRY IT YOURSELF — EXERCISE 7: OBJECTS AND CLASSES

Reinforces: the heap object layout you just saw.

STEP	WHAT TO DO
Goal	Create Person.java with fields, a constructor, and a method; instantiate it in main
File	Person.java (in examples/module-01-exercises/)
Compile	javac Person.java
Run	java Person
Key concept	new Person(...) allocates the object's fields on the heap; person itself is only a reference, held on the stack
Reference	exercises/exercise-07-objects.md

```
$ cd examples/module-01-exercises
$ javac Person.java
$ java Person  // prints: Aman is 21 years old
$ javap -c Person  // putfield stores name/age into the new object on the heap
```

TRY IT NOW Complete Exercise 7 before continuing — see exercises/exercise-07-objects.md.

HEAP VS STACK COMPARISON

Java uses Stack memory for method execution and Heap memory for objects and dynamic data.

FEATURE	STACK	HEAP
`What it stores`	Primitives, references, method calls	Objects and arrays
`Speed`	Very fast	Slower than stack
`Size`	Small	Large
`Memory Allocation`	Static allocation	Dynamic allocation
`Management`	Automatically removed	Garbage collected by JVM
`Lifetime`	Method call ends	Until no references remain
`Examples`	Local variables, return addresses	Objects, Arrays, Collections
`Safety`	Each thread has its own stack	Shared among all threads

KEY TAKEAWAY int x is a primitive in the Stack; String name is a reference in the Stack pointing to a "Java" object in the Heap.

WHY GARBAGE COLLECTION IS NEEDED

Java creates many objects during execution — when they're no longer needed, GC automatically reclaims their memory.

Without GC

- Memory keeps filling up — eventually OutOfMemoryError.

With GC

- Unused memory is reclaimed — application runs smoothly.

WHY GARBAGE COLLECTION IS NEEDED

Prevents Leaks

Releases unreferenced memory.

Improves Performance

Frees memory to run efficiently.

Prevents OOM

Avoids JVM crashes.

Simplifies Dev

No manual memory freeing.

Better Reuse

Memory reused for new objects.

REACHABILITY SPECTRUM Not all "garbage" is the same: Reachable (live, accessible from GC Roots) → Weakly Reachable (only Soft/Weak/Phantom refs — GC-eligible) → Unreachable (orphaned, true garbage). Without GC: memory exhaustion, degraded performance, and unpredictable crashes.

KEY TAKEAWAY How it works, high level: Mark reachable objects from GC Roots → Sweep unreferenced objects → Compact (optional) → memory reused.

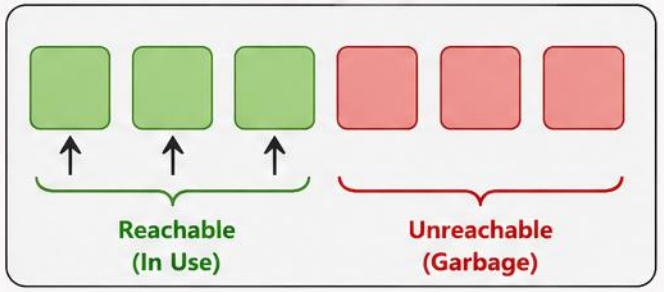
Why Garbage Collection Needed in Java?

In Java, objects are created in heap memory.
When objects are no longer used, they become **unreachable**. Garbage Collection (GC) automatically reclaims that memory, preventing memory leaks and OutOfMemoryError.

Without Garbage Collection

Unreferenced objects remain in memory.
Heap keeps filling up → Memory leak → OutOfMemoryError

Heap Memory



Heap gets full
Application slows down and may crash with **OutOfMemoryError**

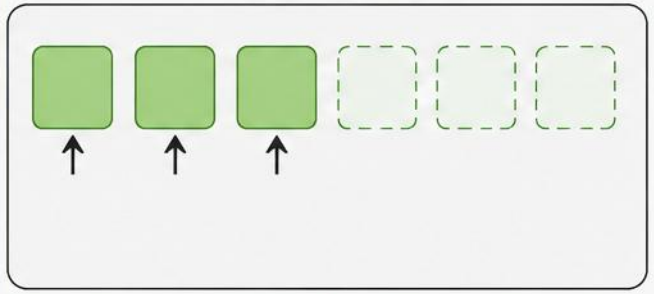
Benefits of Garbage Collection

- **Reclaims memory** occupied by unreachable objects
- **Improves application** performance and stability
- **Prevents memory leaks** and OutOfMemoryError
- **Automatic memory management** – less developer effort

With Garbage Collection

Unreachable objects are detected and memory is reclaimed for future use.

Heap Memory



Memory is freed
Application runs smoothly with optimal memory usage

Live / Reachable Object | Unreachable Object (Garbage) | Reference (In Use) | Freed Memory

Key Takeaway: Garbage Collection is essential in Java to automatically free up memory from unreachable objects, ensuring better performance, reliability, and a better developer experience.

Object Lifecycle in Java

From Creation to Garbage Collection

Key Points

- An object is created in heap memory.
- It remains in use as long as it is reachable.
- When no references point to it, it becomes unreachable.
- Garbage Collector reclaims memory at some later time.
- The timing of garbage collection is not deterministic.

1. Creation

Object is created using **new** keyword.

2. In Use (Reachable)

Object is referenced by variable(s) and can be used.

3. Unreachable

No references point to this object anymore.

4. Eligible for Garbage Collection

JVM identifies unreachable objects as candidates for cleanup.

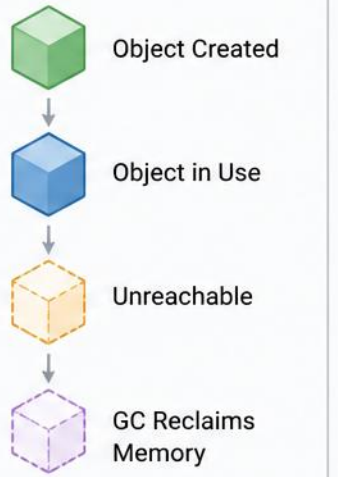
5. Garbage Collected

Memory occupied by the object is reclaimed by Garbage Collector.

Example

```
Student s = new Student();
s.setName("Alice");
s = null; // no more reference
// Eligible for GC
// Garbage Collector reclaims it
// at some later time
```

Heap Memory



Legend

- Active State
- Reachable State
- Unreachable State
- GC Eligibility
- Collected State

Note: Garbage collection in Java is automatic, but not immediate. The JVM decides when and how to reclaim memory.

HOW GARBAGE COLLECTION WORKS IN JAVA

GC identifies objects that are no longer reachable and reclaims their memory for reuse.

GC ROOTS — STARTING POINTS

- Local variables on the Stack, Static variables, Active Threads, JNI References, JVM Internal References.
- GC traverses the object graph from these roots and marks everything reachable; unmarked objects are unreachable and eligible for removal.

KEY TAKEAWAY Some GC algorithms compact the heap afterward, moving live objects together to reduce fragmentation.

COMMON GC ALGORITHMS

Four collectors, each trading off pause time, throughput, and compaction differently.

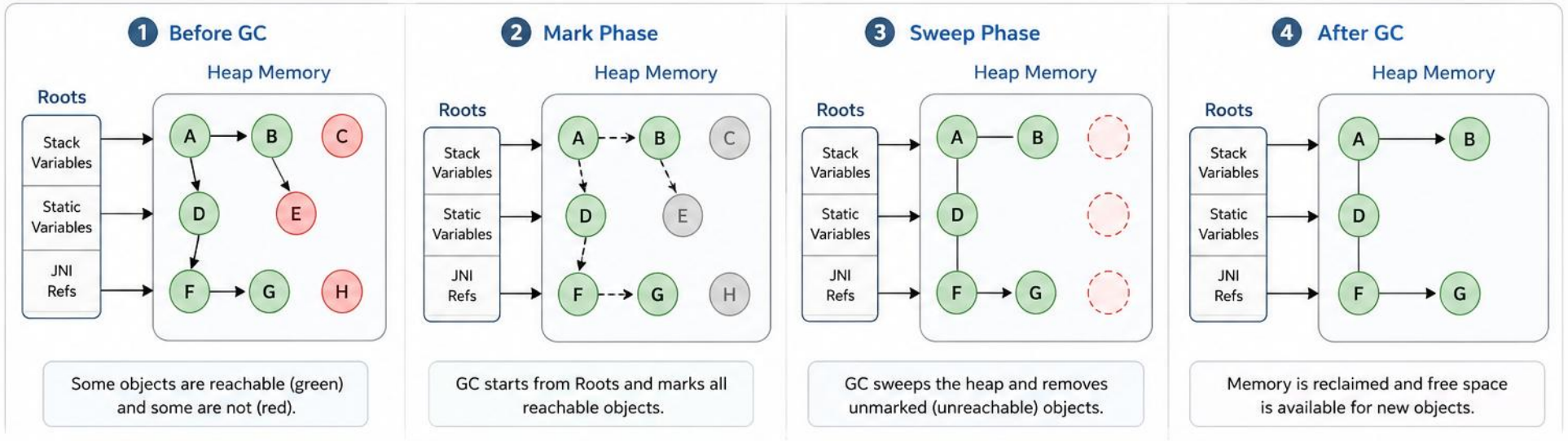
ALGORITHM	TYPE	KEY IDEA	COMPACTION	USE CASE
`Serial GC`	Stop-the-world	Mark-Sweep-Compact	Yes	Small heaps, simple apps
`Parallel GC`	STW, multithreaded	Parallel Mark-Sweep-Compact	Yes	Throughput oriented
`CMS`	Mostly concurrent	Mark-Sweep, no compact	No	Low pause, deprecated
`G1 GC`	Mostly concurrent	Region-based, incremental	Sometimes	Large heaps, balanced

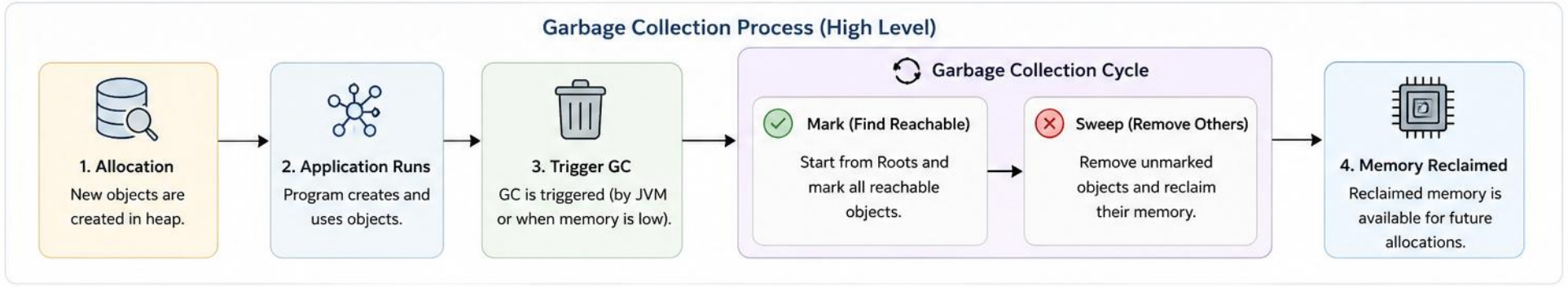
- Useful JVM options: -XX:+UseG1GC, -XX:+PrintGCDetails, -verbose:gc, -Xlog:gc* (JDK 9+).

KEY TAKEAWAY GC runs automatically — developers don't free memory manually, but choosing and tuning the right algorithm still matters.

How Garbage Collection Works in Java

Garbage Collection reclaims memory occupied by objects that are no longer reachable.





GC runs automatically in the background.

Developers don't free memory manually.

Different GC algorithms optimize pause time and throughput.

Java

OVERVIEW OF G1 GARBAGE COLLECTOR

Garbage-First — a server-style collector designed for large heaps, default since Java 9.

Low Pause Times

Targets e.g. 200 ms pause goal.

Region Based

Heap split into 1–32 MB regions.

Garbage First

Collects most-garbage regions first.

Mostly Concurrent

Most work runs with the app.

Scalable

Works well on multi-GB heaps.

HEAP LAYOUT — REGION BASED

THE G1 CYCLE Alternates short pauses with concurrent work: Initial Mark → Root Region Scan → Concurrent Mark → Remark → Cleanup → Mixed Collection → Concurrent Cleanup. Key options: -XX:+UseG1GC, -XX:MaxGCPauseMillis=200, -XX:InitiatingHeapOccupancyPercent=45.

KEY TAKEAWAY G1 divides the heap into many equal-sized regions (a power of 2 between 1 MB and 32 MB, chosen automatically).

G1 GC CYCLE — DETAILS & KEY OPTIONS

What actually happens during each phase, and the flags that control them.

PHASE	TYPE	WHAT HAPPENS
`Initial Mark`	STW	Marks GC Roots and directly reachable objects.
`Root Region Scan`	STW, short	Scans Root regions (e.g. Survivor/Old) for references.
`Concurrent Mark`	Concurrent	Marks live objects across the heap while the app runs.
`Remark`	STW	Finalizes marking; processes reference queues.
`Cleanup`	STW + Concurrent	Reclaims fully-empty regions; updates statistics.
`Mixed Collection`	STW	Evacuates live objects from selected regions to free space.
`Concurrent Cleanup`	Concurrent	Clears remembered sets, prepares for the next collection.

MORE JVM OPTIONS FOR G1

OPTION	PURPOSE
`-XX:G1HeapRegionSize`	Region size, 1M–32M (auto)
`-XX:ParallelGCThreads`	Number of GC worker threads (auto)
`-XX:ConcGCThreads`	Number of concurrent GC threads (auto)

G1 VS OTHER COLLECTORS

How G1 compares against Serial/Parallel and CMS across the dimensions that matter.

ASPECT	SERIAL / PARALLEL	CMS	G1
`Pause Time`	Long	Shorter than Serial/Parallel	Short, predictable (goal-based)
`Throughput`	Highest	Lower (concurrent overhead)	High, tunable
`Compaction`	Yes (STW)	No (fragmentation risk)	Yes, incremental
`Heap Size Support`	Small–Medium	Medium	Large (multi-GB+)
`Tuning Complexity`	Low	High	Moderate
`Default in Java`	Up to 8 (Parallel)	Removed in Java 14	9+ (most cases)

KEY TAKEAWAY G1 became the default collector in Java 9 because it balances throughput and pause time far better than its predecessors at scale.

OVERVIEW OF ZGC

A scalable, low-latency collector — production-ready since JDK 15, aiming for pauses under 10 ms.

Ultra Low Pause

<10 ms, regardless of heap size.

Scalability

Hundreds of MBs to multi-TB heaps.

Cross-Platform

Linux, macOS, Windows, x64, AArch64.

Colored Pointers

Load barriers enable concurrency.

Production Ready

Stable since JDK 15.

HOW ZGC WORKS — HIGH LEVEL

1. **Mark**
2. **Relocate**
3. **Remap**
4. **Reclaim**
5. **Repeat**

KEY TAKEAWAY Most ZGC work is concurrent with application threads — only short pauses handle root marking, remapping and finalization.

ZGC — KEY FEATURES

Six characteristics that let ZGC hold sub-10ms pauses even on multi-terabyte heaps.

Low Pause Times

- <10 ms, independent of heap size.

Concurrent by Design

- Marking, relocating, reclamation all concurrent.

Load Barrier

- Handles concurrent movement/remapping on every load.

Region-Based

- Heap divided into 2 MB regions by default.

Dynamic Resizing

- Grows and shrinks heap with low latency.

Generational (JDK 21+)

- Generational mode improves performance further.

- Key options: -XX:+UseZGC · -Xms / -Xmx · -XX:ZCollectionInterval · -Xlog:gc* (recommended).

KEY TAKEAWAY Use ZGC when every millisecond matters — it is designed for large heaps and latency-sensitive applications.

ZGC VS OTHER COLLECTORS

Where ZGC fits against Serial/Parallel and G1 on latency-critical workloads.

ASPECT	SERIAL / PARALLEL	G1	ZGC
`Pause Time`	Long	Short, predictable	Ultra-low (<10ms)
`Throughput`	Highest	High, tunable	Slightly lower than G1
`Scalability`	Small–Medium heaps	Large heaps	Multi-TB heaps
`Latency Predictability`	Low	Moderate	Very high
`Best Use Case`	Batch jobs, small apps	General-purpose services	Latency-sensitive, huge heaps

MORE JVM OPTIONS FOR ZGC

OPTION	PURPOSE
`-XX:ZUncommitDelay`	Delay before uncommitting unused memory (default 300s)
`-XX:ConcGCThreads`	Number of concurrent GC threads (auto)

Overview of ZGC

ZGC (Z Garbage Collector) is a scalable, low-latency garbage collector designed for modern, large-heap applications.

Key Features

Low Latency
Designed to keep GC pauses typically under 10ms, regardless of heap size.

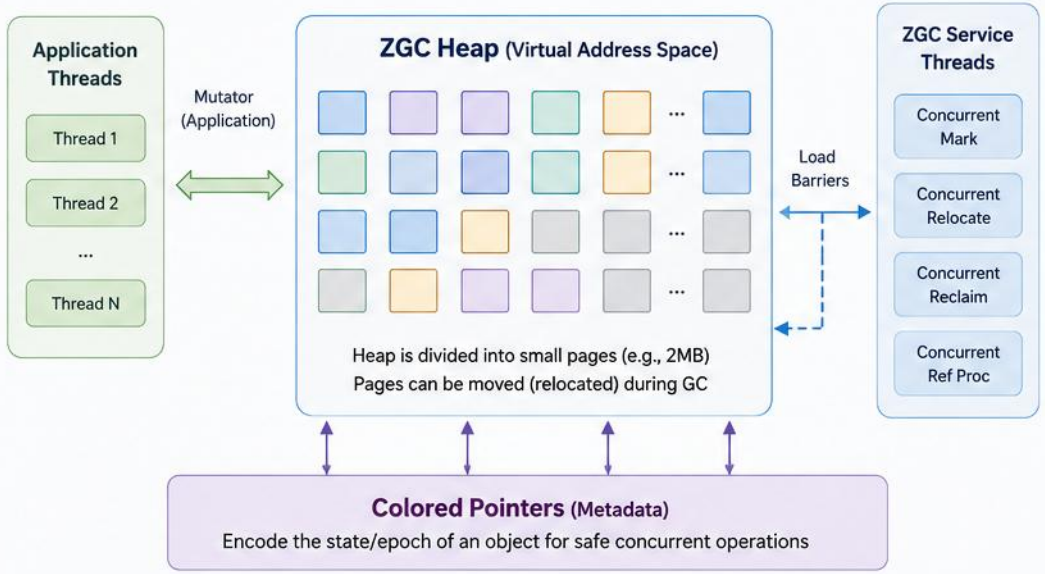
Scalability
Handles multi-terabyte heaps and large numbers of application threads.

Concurrent by Design
Most of the work is done concurrently with the application.

Load Barriers
Uses load barriers and colored pointers to enable concurrent compaction.

High Throughput
Built for production workloads that require both low latency and high throughput.

ZGC Architecture (High Level)



How ZGC Works

1. **Concurrent Mark**
   Finds live objects concurrently while the application runs.
2. **Concurrent Relocate**
   Live objects are copied to new locations concurrently.
3. **Concurrent Reclaim**
   Unused memory is reclaimed concurrently.
4. **Reference Processing**
   References are processed concurrently.
5. **Repeat**
   The cycle repeats to keep latency low.

ZGC Cycle (Simplified)

Benefits

Consistent Low Pauses
Keeps pause times predictable and short, even at large scales.

Large Heap Support
Efficient for heaps from GBs to multiple TBs.

Production Ready
Built for real-world applications with minimal tuning.

Future Proof
Continuously improved with new JDK releases.

In Short: ZGC is a concurrent, scalable, and low-latency garbage collector that enables Java applications to handle large heaps with predictable performance and minimal pause times.

Java

GARBAGE COLLECTION BEST PRACTICES

Good GC performance starts with good code, the right JVM configuration, and continuous monitoring.

THE GOAL

- **Low Pause Times**
- **High Throughput**
- **Efficient Memory**
- **Stable & Reliable**
- **Lower Cost**

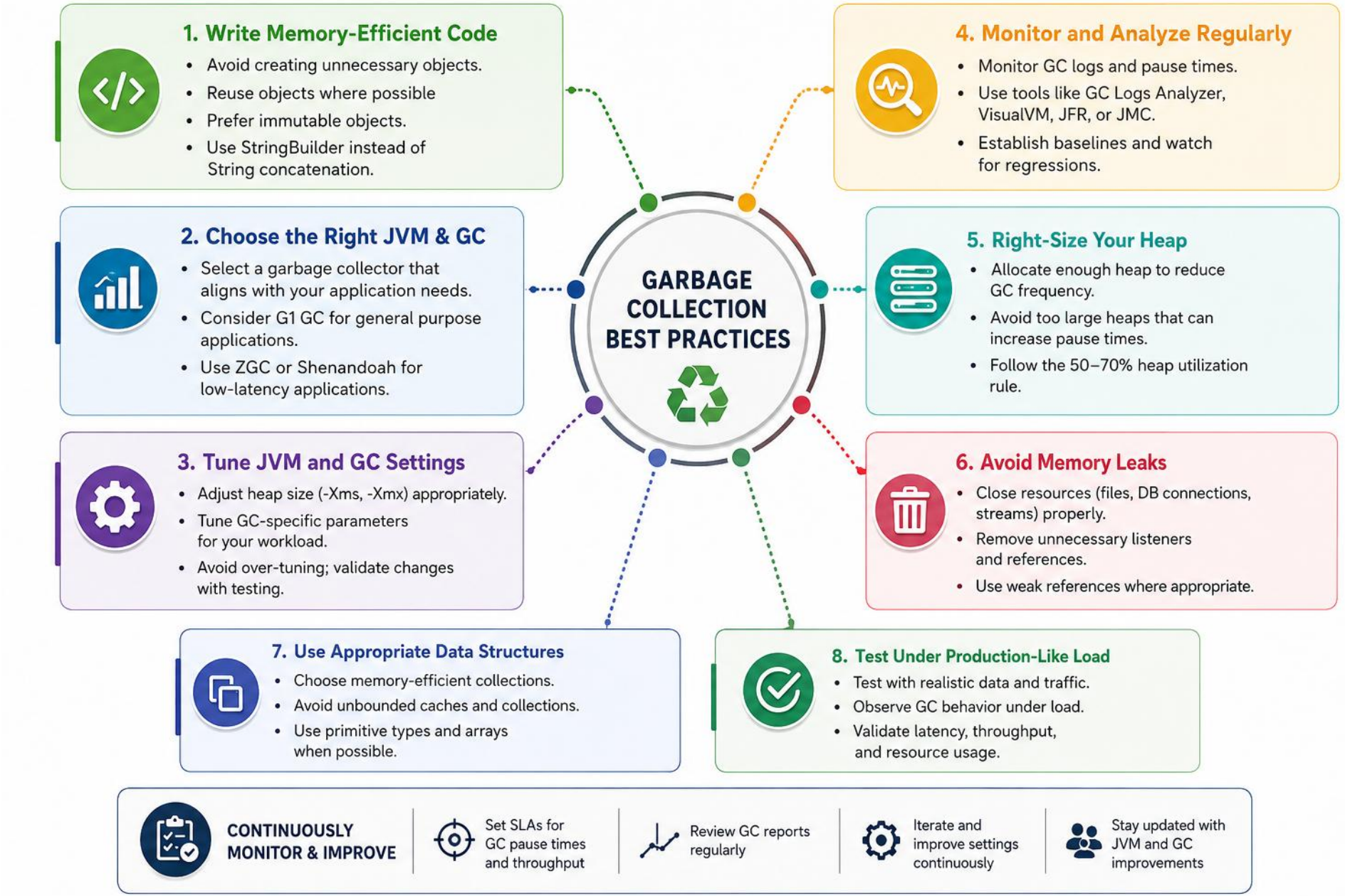
Write Memory-Efficient Code

- Avoid leaks (unclosed resources, static collections).
- Minimize object creation in tight loops; reuse objects.
- Prefer primitives over objects in collections.
- Use try-with-resources and unregister listeners; reach for WeakHashMap / WeakReference for caches instead of hard leaks.

Choose the Right GC

- <10ms latency → ZGC; JDK 8-14 → Shenandoah.
- Balanced → G1 (default, JDK 9+).
- Max throughput → Parallel; simple apps → Serial.
- Tune with -XX:+UseG1GC, -XX:MaxGCPauseMillis=200, -XX:InitiatingHeapOccupancyPercent=45; set -Xms = -Xmx in production.

KEY TAKEAWAY For most applications on JDK 9+, G1 GC is a good general-purpose choice.



JVM STARTUP SEQUENCE

From the java command to your main() method — a well-defined sequence of steps.

1. Load JVM: the JVM process starts.
2. Initialize JVM internal components.
3. Load Bootstrap Class Loader (core classes).
4. Initialize subsystems: Threads, GC, JIT.
5. Parse command-line options (-Xms, -XX...).
6. Set up classpath and module path.
7. Load the main class via the Application Class Loader.
8. Verify bytecode & prepare for execution.
9. Resolve symbolic references of the main class.
10. Invoke main() and start program execution.

KEY TAKEAWAY Most startup time is spent in class loading, verification, and JIT warm-up — keep the classpath clean and lightweight.

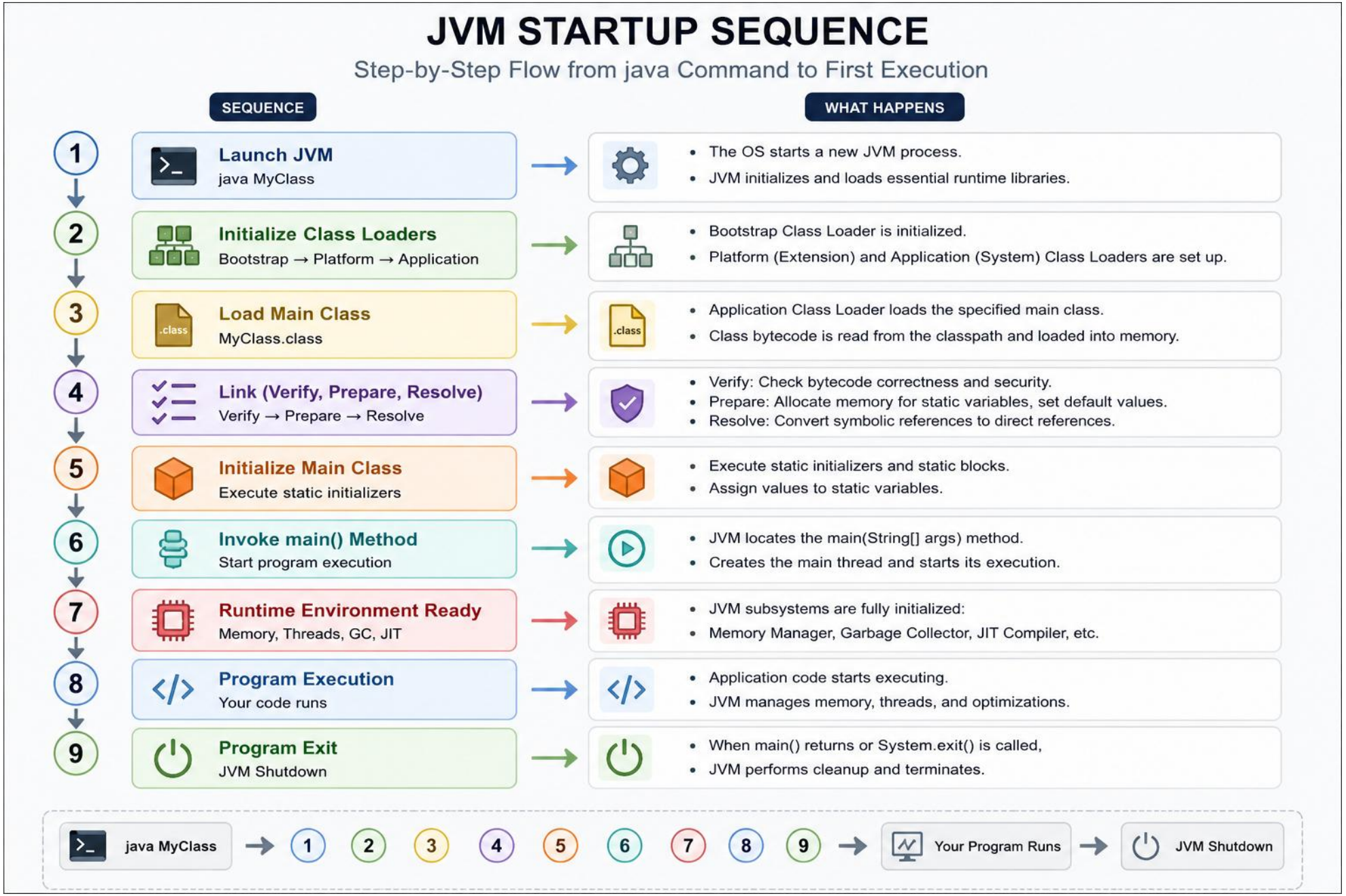
JVM STARTUP — USEFUL OPTIONS & TROUBLESHOOTING

The flags and tools you'll reach for when startup misbehaves.

OPTION	PURPOSE
`-Xms<size> / -Xmx<size>`	Initial / maximum heap size
`-XX:+UseG1GC`	Use the G1 Garbage Collector
`-verbose:class`	Print class-loading info
`-Dkey=value`	Set a system property

- Use -verbose:class to see class loading; -Xlog:gc* (JDK 9+) or -XX:+PrintGCDetails for GC logs.
- Use -XX:+PrintCompilation to see JIT activity; check classpath/module path if the main class isn't found.

KEY TAKEAWAY Command-line options and classpath control startup behavior — bytecode verification ensures safety before main() ever runs.



APPLICATION LIFECYCLE INSIDE JVM

From launch to termination — the JVM loads, runs, manages, and finally terminates the application.

- Start: OS creates the process; JVM is loaded and performs basic setup.
- Load: Class Loaders load required .class files into the Method Area/Metaspace.
- Initialize: Verification, preparation, resolution, and static initialization run.
- Run: main() executes; objects are created, threads run, GC runs in the background.
- Shutdown: shutdown hooks execute, non-daemon threads finish, resources are released.
- Terminate: JVM stops, all memory returns to the OS, process exits with a status code.

KEY TAKEAWAY Shutdown hooks allow graceful cleanup before the JVM process exits.

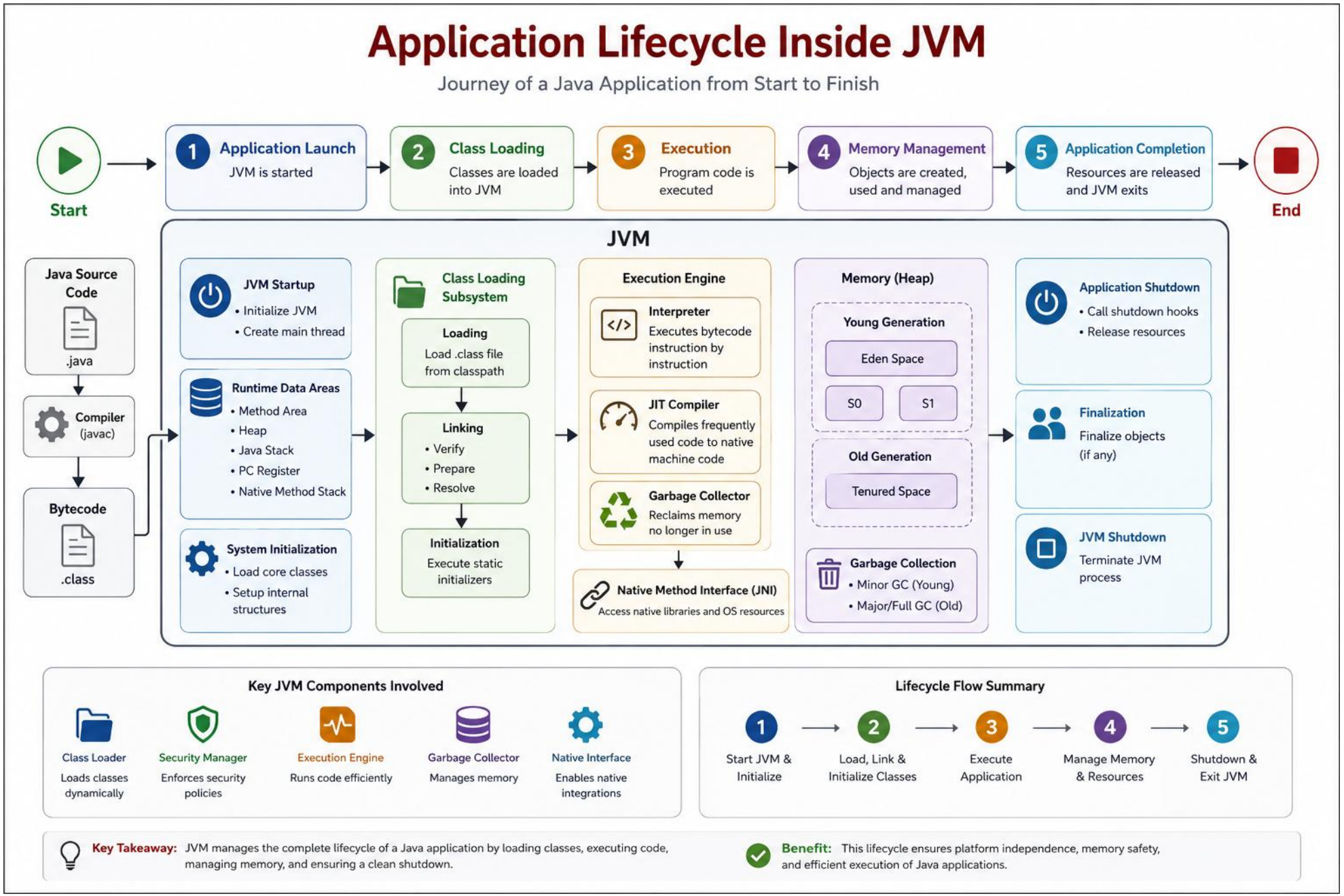
RUNTIME DATA AREAS & APPLICATION STATES

Which memory area is active during each phase of the lifecycle.

AREA	USED IN	PURPOSE
`Method Area (Metaspace)`	Load, Initialize	Class metadata, constant pool, method info.
`Heap`	Run	Stores objects and arrays.
`Java Stacks`	Run	Method call frames and local variables.
`Code Cache`	Run	Stores JIT-compiled code.
`PC Register`	Run	Stores the address of the next instruction per thread.
`Native Method Stack`	Run	Used by native (JNI) methods.

- Application States: New → Loading → Linking → Initialized → Running → Shutting Down → Terminated.

KEY TAKEAWAY Use jcmd, jmap, jstack, VisualVM, or JFR to monitor the JVM during any of these lifecycle phases.



TEN COMMON PERFORMANCE PROBLEMS (1-5)

Symptom, cause, and fix for the issues you'll encounter most often in production.

PROBLEM	SYMPTOMS	CAUSES	IMPACT	HOW TO FIX
`High CPU Usage`	CPU near 100%, slow response	Tight loops, excessive object creation	High latency, poor throughput	Profile with jstack/JFR; optimize code
`Excessive GC`	Frequent/long GC pauses	Too many short-lived objects	High pause time, low throughput	Reduce allocations; right-size heap
`High Memory / OOM`	Gradual growth, OOM crashes	Memory leak, uncleared caches	OOM, crashes, service downtime	Heap dump + MAT; fix the leak
`Thread Contention`	Many WAITING/BLOCKED threads	Synchronization, I/O in locks	Poor scalability, high latency	Analyze jstack; reduce lock scope
`Too Many Threads`	High thread count, context switching	Thread-per-request, unbounded pools	High overhead, low throughput	Use bounded thread pools, async I/O

THE 5 ROOT-CAUSE BUCKETS Almost every performance problem traces back to: CPU & Threads, Memory & GC, I/O & Disk, Application Code, or JVM Configuration. Discipline: identify the symptom → find the root cause → apply the targeted fix — never tune blindly.

KEY TAKEAWAY Five more common performance problems continue on the next slide.

TEN COMMON PERFORMANCE PROBLEMS (6-10)

Symptom, cause, and fix for the issues you'll encounter most often in production.

PROBLEM	SYMPTOMS	CAUSES	IMPACT	HOW TO FIX
`Slow I/O`	High I/O wait	Blocking I/O, slow queries	Increased latency, low throughput	Optimize queries, use async I/O
`Class Loading Issues`	High load time, Metaspace growth	Dynamic loading, classloader leaks	Startup delays, higher memory	Fix leaks, proper delegation
`Inefficient Code`	High CPU/memory, poor throughput	Bad algorithms, excess logging	High resource use, slow app	Profile & optimize hot paths
`Large Heap / Long GC`	Long Full GC pauses	Oversized heap, wrong GC tuning	Long pauses, timeouts	-Xms=-Xmx; tune G1 region/IHOP
`Poor JVM Config`	Unstable performance	Wrong heap/GC/thread settings	Resource waste, unpredictable perf	Review/tune options with real data

KEY TAKEAWAY Tools to identify these: jstat, jstack, jmap, JFR, VisualVM, and GC logs.

PERFORMANCE PROBLEMS — GENERAL BEST PRACTICES

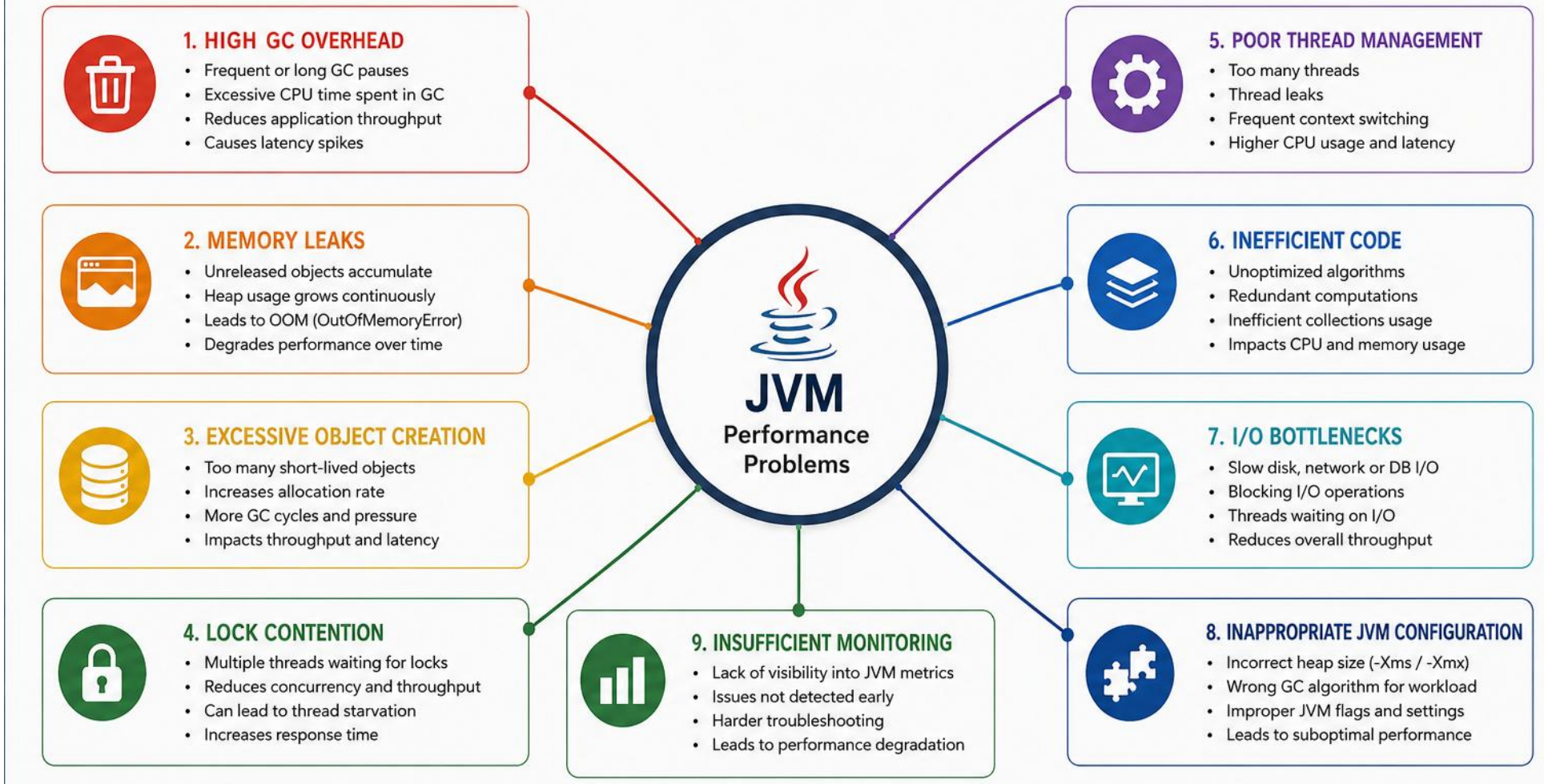
The discipline that catches most issues before they reach production.

- Monitor in production continuously; set alerts on CPU, memory, GC, and threads.
- Understand your workload before you tune it.
- Profile before optimizing — don't guess.
- Change one thing at a time, and test under real load.

KEY TAKEAWAY Measure → Analyze → Fix → Verify → Monitor — the same cycle applies to every category of performance problem.

COMMON JVM PERFORMANCE PROBLEMS

Key issues that impact JVM performance in production systems



OUTOFMEMORYERROR & MEMORY LEAKS

OOME occurs when the JVM cannot allocate memory; leaks happen when unneeded objects stay strongly referenced.

- JVM memory areas that can run out: Heap (objects), Metaspace (class metadata), Stack (threads), Code Cache (JIT), Direct Memory (NIO/off-heap).

MEMORY ERROR TYPE	OCCURS WHEN	TYPICAL CAUSE
`Java heap space`	Heap memory full	Too many objects, memory leaks
`Metaspace`	Metaspace full	Too many classes, classloader leak
`StackOverflowError`	Thread stack full	Deep recursion, too many calls
`Unable to create native thread`	OS cannot create threads	Too many threads, OS limits
`GC overhead limit exceeded`	>98% time in GC, <2% heap recovered	Memory leak, or heap too small
`Direct buffer memory`	Direct/off-heap memory full	NIO direct buffers, Netty/Unsafe usage

KEY TAKEAWAY Different OOME types point to different exhausted memory areas — the error text tells you where to look first.

MEMORY LEAKS — CAUSES & DIAGNOSIS

Objects that are no longer needed but remain strongly reachable cause gradual memory growth.

COMMON CAUSES

- Collections never cleared; unbounded caches with no eviction policy.
- Listeners/callbacks registered but never removed.
- ThreadLocal misuse, especially in thread pools.
- ClassLoader leaks and unnecessary static references.

HOW TO DIAGNOSE

1. Monitor heap usage, GC activity, thread count.
2. Take a heap dump: jmap -dump:live,format=b,...
3. Analyze with Eclipse MAT / VisualVM / YourKit.
4. Check GC logs for frequent GC and long pauses.

HOW TO PREVENT LEAKS Design for bounded memory (cap caches/queues/collections); always clear collections and remove listeners; use weak/soft references and try-with-resources; call ThreadLocal.remove() in thread pools. Increasing heap size is NOT a fix for a leak — it only delays the crash.

KEY TAKEAWAY Quick check: memory keeps rising, Full GC runs but memory doesn't drop, heap dump shows many objects that should be dead — likely a leak.

Out of Memory Error and Memory Leaks

Understanding the causes, behavior and prevention strategies

1. OUT OF MEMORY ERROR (OOM)

Occurs when the JVM cannot allocate enough memory for an object or processing.

java.lang.OutOfMemoryError
Application crashes or becomes unstable

Where OOM Happens

Heap	Metaspace	Stack	Direct Memory
OutOfMemoryError: Java heap space	OutOfMemoryError: Metaspace	OutOfMemoryError: unable to create new native thread	OutOfMemoryError: Direct buffer memory

Typical OOM Example (Heap)

```
List<int[]> list = new ArrayList<>();
while (true) {
  list.add(new int[1024 * 1024]); // 1 MB
}
// Eventually throws:
// java.lang.OutOfMemoryError: Java heap space
```

Heap Exhaustion Over Time

How to Prevent OOM

- **Right-size the heap (-Xmx and -Xms)**
  Allocate appropriate memory for the application.
- **Optimize memory usage**
  Avoid creating unnecessary objects and large collections.
- **Use efficient data structures & caching**
  Prefer streaming, paging, and lazy loading.
- **Monitor memory**
  Use tools like VisualVM, JFR, GC logs to identify issues.
- **Release resources**
  Close streams, connections, and other resources.
- **Avoid memory fragmentation**
  Tune GC settings for better performance.
- **Load test your application**
  Simulate real workload to find memory limits early.

2. MEMORY LEAKS

Occurs when objects that are **no longer needed are still referenced (directly or indirectly)** so GC cannot reclaim memory.

Memory Leak Impact

- Continuous memory growth
- Performance degradation
- Eventually leads to OOM

Common Causes of Memory Leaks

Static Collections
Objects added to **static** collections are never released.

```
static List<User> users = new ArrayList<>();
users.add(new User(...));
```

Unreleased Event Listeners / Callbacks
Listeners registered but never unregistered keep references.

```
button.addActionListener(this);
// forgot to removeActionListener(this);
```

Improper Cache Usage
Cache grows without bounds or entries are never evicted.

```
Map<String, Object> cache = new HashMap<>();
cache.put(key, value); // keeps growing
```

Long-Lived Threads
Threads holding references prevent objects from being GC'ed.

```
new Thread(() -> {
   while(true) { /* holds reference */ }
 }).start();
```

Unclosed Resources
Files, DB connections, streams not closed retain memory.

```
InputStream is = new FileInputStream(file);
// not closed
```

How to Prevent Memory Leaks

- **Remove references when done**
  Set references to **null** or remove from collections.
- **Unregister listeners / callbacks**
  Always remove listeners in **finally** blocks or close methods.
- **Use bounded caches**
  Use size-limited caches with expiration (e.g., Caffeine, Guava Cache).
- **Close resources properly**
  Use **try-with-resources** for streams, connections, and sockets.
- **Monitor & analyze**
  Use **Heap Dump**, MAT, VisualVM to detect and fix leaks.

Tools to Detect & Diagnose

GC Logs · VisualVM · JFR (Java Flight Recorder) · Heap Dump · MAT (Memory Analyzer Tool) · JMC (Java Mission Control)

JVM MONITORING BASICS

Observe – Understand – Optimize. Monitoring helps you diagnose issues and optimize performance in production.

App Health

Better Performance

Faster Diagnosis

Capacity Planning

Prevent Outages

WHAT CAN WE MONITOR?

- CPU Usage, Memory Usage (Heap, Non-Heap, Metaspace, Direct Memory).
- Garbage Collection activity, pause time, and frequency.
- Threads: live, blocked, and deadlocks; Class Loading: loaded/unloaded classes.
- Application Metrics: throughput, latency, errors, custom metrics.
- I/O & Network: disk I/O, network throughput.

THE PIPELINE Application → Metrics Collection → Monitoring System (Prometheus/Datadog) → Dashboards & alerts.

KEY TAKEAWAY Consistent monitoring gives visibility into JVM behavior and detects problems before they become outages.

JDK MONITORING TOOLS (BUILT-IN)

Seven tools that ship with every JDK, no extra installation required.

TOOL	USE	EXAMPLE
`jps`	List running JVM processes	jps -l
`jstat`	GC, memory, class statistics	jstat -gcutil <pid> 5s
`jmap`	Heap dump, memory maps	jmap -dump:live,format=b,file=heap.hprof <pid>
`jstack`	Thread dump (stack traces)	jstack <pid> > thread.txt
`jcmd`	Multipurpose diagnostics (JDK 9+)	jcmd <pid> GC.heap_info
`jinfo`	JVM configuration & system properties	jinfo -flags <pid>
`VisualVM`	Visual monitoring/profiling	visualvm

KEY METRICS TO WATCH Heap Used consistently >70-80%, GC pauses growing longer/more frequent, a steadily climbing thread count, growing Metaspace (leak sign), rising p95/p99 latency or dropping throughput — baseline "normal" first, then watch for deviations.

KEY TAKEAWAY Run these tools as the same user that owns the JVM process you're inspecting.

JVM MONITORING BASICS

Monitor JVM to ensure performance, stability and optimal resource usage

What is JVM Monitoring?

JVM Monitoring is the process of collecting and analyzing data from the JVM to understand its health, performance and resource usage.

Why Monitor JVM?

- Detect performance bottlenecks
- Prevent OutOfMemoryError
- Optimize resource utilization
- Improve application stability
- Understand behavior under load
- Faster issue detection & resolution

Key Areas to Monitor

- Memory Usage (Heap & Non-Heap)
- Garbage Collection (GC)
- Thread Activity
- CPU Usage
- Class Loading
- I/O & Network
- Application Throughput & Latency
- System Resources (OS)

JVM PROCESS

Your Application

JVM RUNTIME

- Heap Memory
- Non-Heap (Metaspace)
- Threads
- Class Loader
- Garbage Collector

Key Metrics

Memory

- Heap Used / Committed
- Heap Max
- Non-Heap Usage
- Metaspace Usage

Garbage Collection

- GC Count
- GC Time
- Pause Time
- GC Throughput

Threads

- Live Threads
- Daemon Threads
- Blocked Threads
- Thread States

CPU

- CPU Usage %
- Load Average
- Application CPU Time

Others

- Class Loading Count
- File Descriptors
- Network I/O
- Application Throughput / Latency

Common Tools

JConsole (Built-in)
GUI tool included in JDK for basic monitoring.

VisualVM (Free)
Powerful visual tool for monitoring & profiling.

JMC (Java Mission Control)
Advanced monitoring, alerts & diagnostics.

Prometheus + Grafana
Open-source monitoring & visualization stack.

Datadog / New Relic / Dynatrace
APM tools for enterprise monitoring.

Command Line Tools
jstat, jmap, jstack, jcmd, jinfo, jps

Monitoring Data Flow

JVM Process (Running Application) → Metrics Collection (JMX / JVMTI / Agents / Tools) → Storage (Time Series DB / Monitoring Backend) → Visualization & Alerts (Dashboards / Alerts / Reports)

Best Practices

- Monitor in all environments (Dev, Test, Prod)
- Set baseline metrics & thresholds
- Use alerts for early detection
- Regularly review GC & memory trends
- Correlate JVM metrics with application & OS metrics

INSTRUCTOR DEMO: COMPILATION AND BYTECODE

From Java source code to JVM instructions — a live walkthrough.

- Write Java source with a main() method and a helper add() method.
- Compile: javac generates Hello.class — syntax checked, semantics analyzed, bytecode generated.
- Run: Class Loader loads Hello.class, Bytecode Verifier checks safety, Execution Engine runs it, JIT optimizes hot code.
- Platform independence: the same Hello.class runs on Windows, Linux, and macOS JVMs and produces the same output.
- Bytecode anatomy: bipush → istore/iload → getstatic → invokestatic/invokevirtual → iadd/ireturn — modern javac compiles string concatenation via invokedynamic, not repeated StringBuilder calls.

DEMO COMMANDS javac Hello.java → javap -c Hello (inspect bytecode) → java Hello (run and compare output).

KEY TAKEAWAY javac compiles .java → .class; the JVM loads, verifies, and executes that bytecode on any platform.

TRY IT YOURSELF — EXERCISE 8: INSPECT BYTECODE

Reinforces: the bytecode anatomy you just walked through.

STEP	WHAT TO DO
Goal	Disassemble Person (or Hello) with javap and note three bytecode instructions
File	Person.class (already compiled in Exercise 7)
Command	javap -c Person
Look for	aload_0, putfield, invokespecial in the constructor; getfield in display()
Key concept	javac compiles once; javap only shows the bytecode; java is what actually runs it
Reference	exercises/exercise-08-javap.md

```
$ cd examples/module-01-exercises
$ javap -c Person
$ javap -c -v Person  // verbose: also shows the constant pool
$ java Person  // confirms the disassembled bytecode is what actually executed
```

TRY IT NOW Complete Exercise 8 before continuing — see exercises/exercise-08-javap.md.

INSTRUCTOR DEMO: MEMORY ALLOCATION

See how objects are created and stored in JVM memory, step by step.

1. JVM loads class HelloMemory; metadata goes to the Method Area.
2. main() starts; a Stack frame is created.
3. new HelloMemory(101, "Alice") is executed.
4. JVM allocates memory for the object on the Heap.
5. Reference obj is stored in the Stack frame.
6. Object fields id and name are stored in the Heap.

KEY TAKEAWAY The Stack holds references. The Heap holds actual objects.

MEMORY AREAS USED & END OF MAIN()

What happens to the object once its method frame is gone.

AREA	USED FOR	IN THIS EXAMPLE
`Method Area`	Class metadata, static variables	Stores HelloMemory class info
`Heap`	Objects and arrays	Stores HelloMemory object and String "Alice"
`Java Stack`	Method calls, local variables, references	Stores reference obj in main() frame
`PC Register`	Address of the next instruction	Not shown directly in this example
`Native Method Stack`	Used by native (JNI) methods	Not used in this example

LIVE MEMORY VIEW (jvisualvm) — HEAP HISTOGRAM

CLASS	INSTANCES	SIZE
`HelloMemory`	1	24 B
`java.lang.String`	1	24 B
`char[]`	6	32 B
`Total`	8	80 B

- When main() completes, its Stack frame is popped and the reference obj is gone.
- If no other references exist, the object on the Heap becomes unreachable and the GC reclaims it later.

KEY TAKEAWAY Hands-on idea: create objects in a loop and watch Heap usage grow in VisualVM, then set obj = null and observe reclamation.

LAB OVERVIEW — JVM AND COMPILATION

Build HelloWorld, Calculator, Employee and MemoryDemo — the Northstar onboarding checklist proving you understand how the JVM compiles, loads, and executes Java code.

- Objectives: write and compile HelloWorld, Calculator, Employee and MemoryDemo; understand compilation; inspect bytecode with javap.
- Run Java programs on the JVM, understand class loading and execution flow, verify platform independence.
- Prerequisites: Lab 0 complete, JDK 21 installed (javac, java, javap), VS Code or IntelliJ IDEA.

KEY TAKEAWAY Lab 1 builds JVM literacy before Spring Boot, Maven, or the Customer Management Platform: source → bytecode → JVM load → execute.

PRE-LAB EXERCISES & SUCCESS CRITERIA

Eight pre-lab exercises to complete before Lab 1, from Hello World to inspecting bytecode.

#	EXERCISE	FOCUS AREA	KEY COMMANDS
1	Hello World	Basic program & execution	javac Hello.java; java Hello
2	Platform Independence (WORA)	Run bytecode without recompiling	java Hello
3	Control Flow	if, for, while, switch	javac ControlFlow.java; java ControlFlow
4	Watch Class Loading	Observe JVM class loading	java -verbose:class Hello
5	Variables and Data Types	Local variables, types	javac Variables.java
6	Methods and Parameters	Method calls, stack usage	javac Methods.java
7	Objects and Classes	Object creation, fields	javac Person.java
8	Inspect Bytecode	Bytecode structure	javap -c Person

- Success criteria: compile without errors, inspect and explain bytecode, run programs and explain output, understand class loading and execution.

KEY TAKEAWAY Try -Xint and -Xcomp to compare interpreted vs. compiled execution, and -verbose:class to watch class loading live.

LAB 1 TASKS AND DELIVERABLES

Write, compile, inspect, run, analyze, and submit — building HelloWorld, Calculator, Employee, and MemoryDemo.

#	TASK	DESCRIPTION	TOOLS
1	Hello World	Print "Hello, JVM!" and run it.	javac; java
2	Inspect Bytecode (javap)	Disassemble HelloWorld.class; read getstatic / ldc / invokevirtual.	javap -c
3	Calculator	add(10, 20) → Sum = 30; observe stack frames and iadd.	javac; java; javap -c
4	Employee (Heap Objects)	new Employee(101, "Aman"); map stack reference to heap object.	javac; java
5	Observe Class Loading	Trace JDK and Employee classes loading with -verbose:class.	java -verbose:class
6	MemoryDemo	Allocate 100,000 Employees; discuss heap pressure and -Xmx.	javac; java
7	Clean & Recompile	Delete *.class, rebuild all four programs, re-verify output.	javac (rebuild)

KEY TAKEAWAY Each task builds toward the Northstar onboarding checklist: four working programs, bytecode literacy, and a clear stack-vs-heap mental model.

DELIVERABLES & EVALUATION CRITERIA

What to submit, and how it's scored.

- Deliverables: four source files (HelloWorld, Calculator, Employee, MemoryDemo), compile/run screenshots, javap -c bytecode capture, class-loading evidence, seven short lab-report answers, and a pushed personal GitHub repo.

EVALUATION CRITERIA	WEIGHT
`Environment Readiness`	10%
`Compile, Run & Inspect Bytecode`	40%
`Heap Model & Class Loading`	20%
`MemoryDemo, Heap Flags & Recompile`	15%
`Evidence, Analysis & Failure Experiments`	15%

KEY TAKEAWAY Compile before running, fix all errors first, and capture clean evidence for all four programs before submitting.

ANALYSIS QUESTIONS (ANSWER IN REPORT)

Seven questions every Lab 1 report must answer, tying HelloWorld, Calculator, Employee and MemoryDemo back to the concepts.

1. What does javac do, and what file does it produce?
2. What is bytecode, and why is it not machine code?
3. Why is bytecode platform-independent?
4. What is the role of the JVM in executing your program?
5. Where are objects stored?
6. Where are method calls and stack frames stored?
7. What happens when a class is loaded by the JVM?

KEY TAKEAWAY These questions map directly to the module's core topics — compilation, bytecode, class loading, and stack memory.

MODULE 1 SUMMARY

JVM, compilation, and program execution fundamentals — the full journey from code to output.

- The JVM is the runtime engine that loads and executes Java bytecode, providing platform independence.
- javac compiles .java source into platform-independent .class bytecode.
- The JVM loads classes, verifies bytecode, manages memory, and executes code starting from main().
- Local variables and method calls live on the Stack; objects are created in the Heap.

KEY TAKEAWAY Write → Compile → Run → Repeat. Bytecode is the bridge between your source code and the JVM.

KEY CONCEPTS & WHAT YOU CAN DO NOW

A one-page glossary, plus the skills you're walking away with.

CONCEPT	MEANING
`Source Code`	Human-readable .java files
`Compilation`	javac converts .java into .class
`Bytecode`	.class files containing JVM instructions
`JVM Execution`	Loads, verifies, and executes bytecode
`Stack / Heap`	Method frames & locals / objects and arrays
`Platform Independence`	Same .class runs on any JVM

- You can now write and compile Java programs confidently, inspect bytecode with javap, and explain how the JVM loads and executes your code.

KEY TAKEAWAY You've built a strong foundation for mastering the JVM — memory, GC, and performance are next.

MODULE 1 KNOWLEDGE CHECK — MULTIPLE CHOICE (1–6)

Choose the best answer. Correct answers are marked ✓.

1. What is the primary role of the JVM?

A) To edit and write Java source code B) To compile Java code into machine code **C) To load, verify and execute Java bytecode ✓** D) To store Java source files

2. Which command compiles a Java source file?

A) java Hello.java **B) javac Hello.java ✓** C) jvmc Hello.java D) compile Hello.java

3. What file is generated after successful compilation?

A) Hello.exe **B) Hello.class ✓** C) Hello.out D) Hello.bin

4. What is the format of JVM bytecode?

A) Platform-dependent machine code B) Human-readable text **C) Platform-independent binary instructions ✓** D) High-level source code

5. Which component loads the .class file into the JVM?

A) Execution Engine **B) Class Loader ✓** C) Bytecode Verifier D) JIT Compiler

6. Where are local variables and method call info stored?

A) Heap B) Method Area **C) Java Stack ✓** D) PC Register

KEY TAKEAWAY Six more multiple-choice questions continue on the next slide.

MODULE 1 KNOWLEDGE CHECK — MULTIPLE CHOICE (7–12)

Choose the best answer. Correct answers are marked ✓.

7. Where are objects and arrays stored in the JVM?

A) Java Stack **B) Heap ✓** C) Method Area D) PC Register

8. What does the Bytecode Verifier do?

A) Optimizes bytecode for faster execution B) Converts bytecode to machine code **C) Ensures bytecode is safe and follows JVM rules ✓** D) Manages memory allocation in heap

9. From which method does the JVM start execution?

A) start() **B) main(String[] args) ✓** C) run() D) execute()

10. Which tool views bytecode instructions of a .class file?

A) java -cp **B) javap ✓** C) javac -d D) jdb

11. Which is true about the same .class file?

A) It runs only on the OS where compiled B) It must be recompiled for each platform **C) It can run on any platform with a JVM ✓** D) It is executed directly by the OS

12. Which component interprets or JIT compiles and executes bytecode?

A) Class Loader **B) Execution Engine ✓** C) Native Libraries D) Java Stack

KEY TAKEAWAY 12 questions, testing the full span of this module — from JVM architecture to class loading to bytecode.

MODULE 1 KNOWLEDGE CHECK — TRUE/FALSE & FILL IN THE BLANKS

Sections B and C, with answers shown.

SECTION B — TRUE OR FALSE

1. Java source code (.java) is human-readable. **True ✓**
2. Bytecode is platform-dependent. **False ✓**
3. The JVM uses the Class Loader to load .class files. **True ✓**
4. Method calls and local variables are stored on the stack. **True ✓**
5. The same .class file can run on Windows and Linux with a JVM. **True ✓**

SECTION C — FILL IN THE BLANKS

1. Converting .java to .class is called → **Compilation**
2. The area in JVM where objects are stored is called → **Heap**
3. The JVM component that verifies bytecode for safety is the → **Bytecode Verifier**
4. The command to disassemble a .class file is → **javap**
5. Local variables are stored in the → **Stack area**

KEY TAKEAWAY Bonus question: why is Java bytecode called platform-independent? Because the same .class file runs on any JVM, regardless of OS.

MODULE 1 KEY TAKEAWAYS (PART 1)

You explored how Java code is compiled to bytecode and executed by the JVM.

- JVM is the Runtime Engine: loads, verifies and executes Java bytecode, providing platform independence.
- Compilation: Source to Bytecode — javac converts .java into platform-neutral .class bytecode.
- JVM Architecture: Class Loader, Bytecode Verifier, Execution Engine, Runtime Data Areas, JNI, Native Libraries.
- Class Loading Process: Loading → Linking → Initialization, via the Delegation Model.
- Memory Management: Heap for objects, Stack for method frames, local variables and references.

KEY TAKEAWAY Execution starts from main(): JVM looks for public static void main(String[] args) as the entry point.

MODULE 1 KEY TAKEAWAYS (PART 2)

Why these fundamentals matter for everything that comes next.

- Bytecode Verification Ensures Safety: the Verifier checks bytecode follows JVM rules, preventing unsafe or malicious operations.
- Key Tools: javac (compiler), javap (disassembler), java (JVM launcher).
- Same .class, Different Platforms: once compiled, the same bytecode runs on Windows, Linux, macOS, and more.
- Stack Stores Method Execution Data: each call creates a frame holding local variables, operand stack, and return info.
- Heap Stores Objects and Arrays: created with new, and reclaimed automatically by the Garbage Collector.

Performance

Tune with JVM internals knowledge.

Debugging

Class loading, stack, heap, bytecode.

Security

Verification protects unsafe code.

Portability

Platform independence, everywhere.

Foundation

Base for memory, GC, concurrency.

KEY TAKEAWAY Write Java code in .java files. Compile with javac. The JVM loads, verifies and executes bytecode. Understand the flow to build better, faster applications.

Great work

You've built a strong foundation for mastering the JVM — how Java code compiles, loads, and runs.

Memory management and performance tuning are next in your toolkit.